AKADEMIKA

FCL-03

FIBER OPTIC ANALOG AND DIGITAL MODULATION & DEMODULATION KIT

EXPERIMENTAL & SERVICE MANUAL

…. A MESSAGE FROM

AKADEMIKA

Today's system designers are faced with tomorrow's problems. Basic electronics is one of the important subject need to teach while learning electronics.
It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.
As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook for any further help.

After unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In **FIBER OPTIC COMMUNICATION LAB** modules, signals are to be monitored with an oscilloscope as explained in the manual. The scope input channel should be ac coupled, unless otherwise indicated. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

TECHNICAL SPECIFICATIONS

Transmitter	**:** 1 No. Siemens Fiber Optic LED Peak wavelength of emission 660 nm visible (SFH756V)
Receiver	**:** 2 Nos. Siemens Photo Detector
Receiver 01	**:** Photo Detector with TTL logic output (SFH551V)
Receiver 02	**:** Photo Diode with responsivity of 0.3 μA/μW (SFH250V)
Modulation Techniques	**:** **i)** Pulse Width Modulation (PWM with variable clock 4KHz, 8KHz, 16KHz, 32KHz) **ii)** Pulse Position Modulation (PPM with variable clock 4KHz, 8KHz, 16KHz, 32KHz)
Multiplexing	**:** 2 channel FDM
Frequency	**:** 1KHz, 2KHz
Amplitude	**:** 0 to 4 Vpp
Frequency Modulator 1	**:** Variable Carrier Frequency from 1 to 15 KHz
Frequency Modulator 2	**:** Variable Carrier Frequency from 1 to 30 KHz
Mux clock	**:** 256KHz
Band Pass Filter 1 & 3	**:** Frequency range 8 to 12 KHz with fc of 10KHz
Band Pass Filter 2 & 4	**:** Frequency range 18 TO 22 KHz with fc of 20KHz
FM demodulator	**:** 2 Nos. PLL detectors **i)** Freq. 1 to 15KHz input signal upto 4Vpp **ii)** Freq. 1 to 30KHz input signal upto 4Vpp
Filters	**:** 2 Nos. 4^{th} Order Butter worth Filters with 3.4KHz cut-off frequency

Fiber Optic Cable
Type : Plastic Fiber
Step Index, Multimode
Core Refractive Index -n1 : 1.492
Clad Refractive Index -n2 : 1.406
Numerical Aperture : 0. 50
Acceptance Angle : 60 degrees
Fiber Diameter : 1000 microns
Outer diameter : 2.2 mm

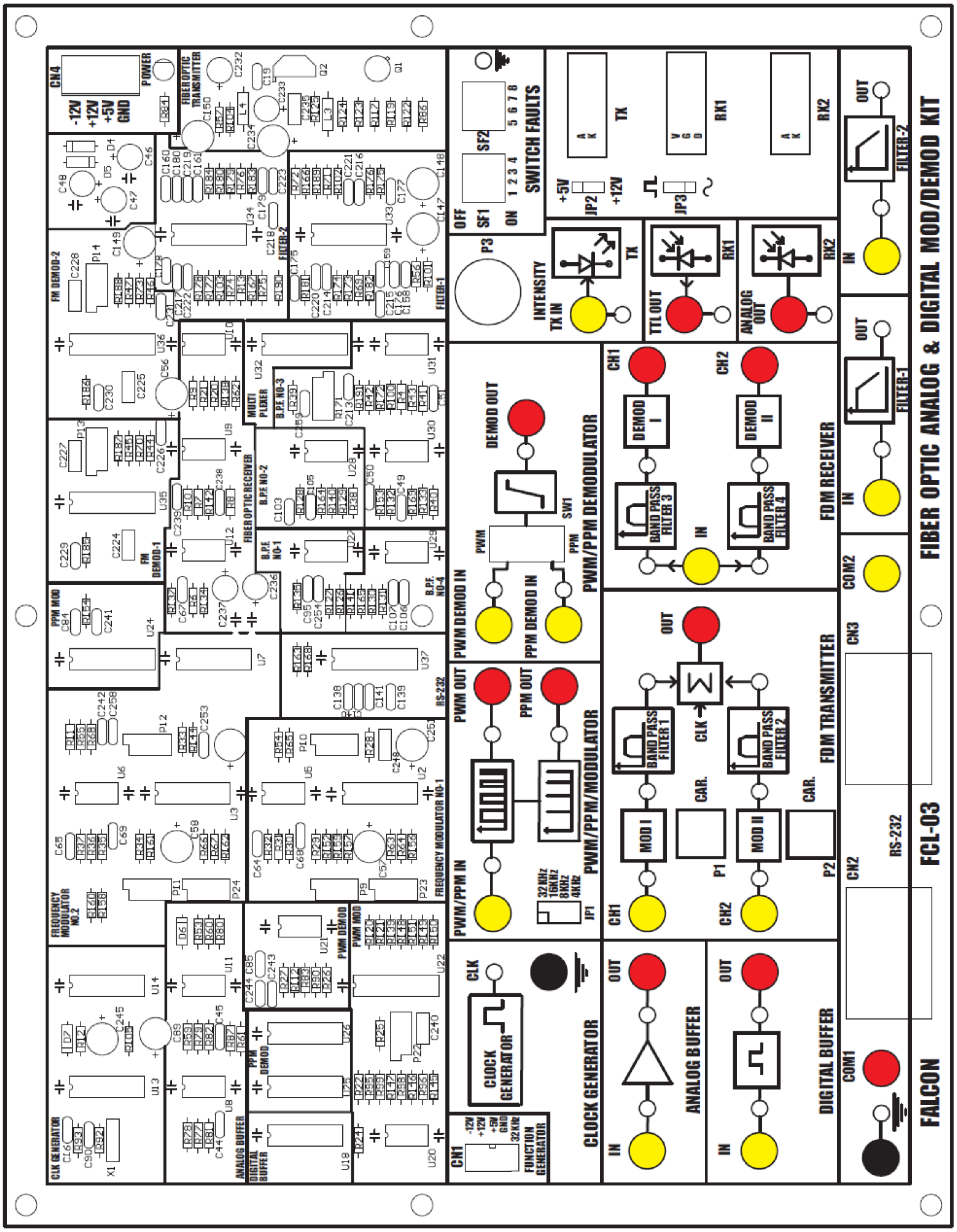
Fiber lengths : 1Meters

Power Supply : GND, +5V, +12V, -12V

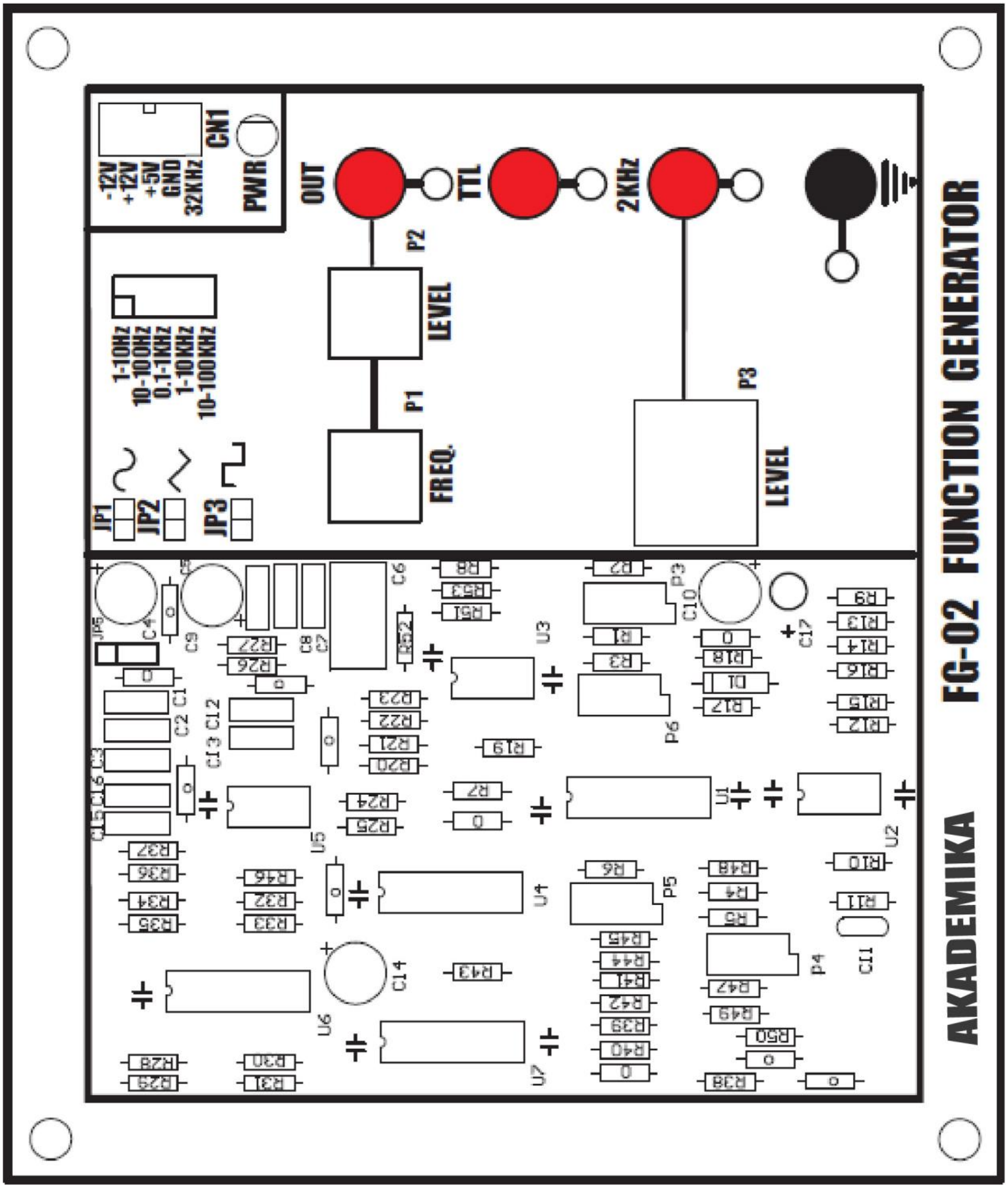
Deliverables

Mini Links	: 07
Short Links	: 02
FCL-03 EXP & SER Manual	: 01
Power Supply	: 01
Power Chord	: 01
Plastic fiber Step Index, Multimode, 1000um,1meter	: 01
Jumper caps small (5mm)	: 03

FUNCTIONAL BLOCK
FCL-03



FG-02



INTRODUCTION

FIBER OPTICS TECHNOLOGY is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T, sprint have saturated the airways & print media with advertisements heralding this bright & new technology, Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

And their enthusiasm is valid. Today medical fiber optic systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologies ever come close to the fantastic growth rate that market analyst love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will prevade our homes, workplaces & recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronic newspapers become feasible. Two-way fiber optic will allow you to "attend college" in the comfort of your home.

Anyone who takes an interest in fiber optic & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few " experts" exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons exciting with the prospect of making new discoveries finding new applications for the technology.

This kit is an introduction to fiber optics communication technology for instructors, students & hobbyist. Welcome to the fascinating & expanding world of fiber optics. We hope that you find the kit challenging, stimulating & - yes – even fun.

ABOUT FIBER OPTICS

Before fiber optics came along, the primary means of real time data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic (radio) waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances – using light energy. Although initially a very controversial technology fiber optics has today been shown to be very reliable & cost effective
Light as utilized for communication has a major advantages because it can be manipulated (modulated) at significant higher frequencies than electrical signals can. For example, a fiber optic cable can carry up to 100 million times more information

than telephone line! The fiber optic cable has lower energy loss & wider bandwidth capabilities than copper wire.

As you will learn, fiber optic communication is a quite simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today. Fiber optics became reality when several technologies came together at once. It was not an immediate process, nor was it easy, but it was most impressive when it occurred. An example of one critical product, which emerged from that technological merger, was the semiconductor LED that we used in our kit.

How Optical Fiber Works?

The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Air & water are a case in point as shown in fig. 1. The light ray traveling through air actually is bent as it enters the water. The amount of bending depends on the refractive indices of the two materials involved & also on the angle of the incoming (incident) ray of light as it strikes the interface. The angle of the incident ray is measured from a line drawn perpendicular to the surface. The same is true for the angle of the outgoing (transmitted) ray of light after it has been bent.

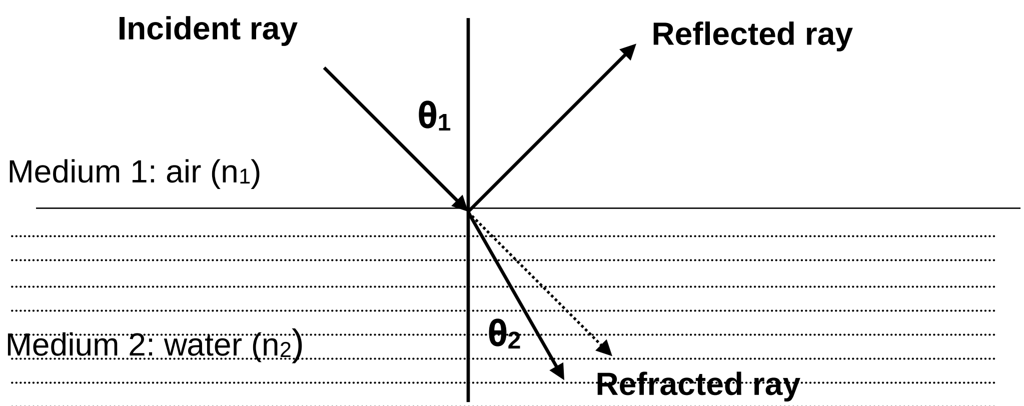


FIG. 1: The different portions of a light ray at a material interface.

The mathematical relationship between the incident ray & the refracted ray is explained by **Snell's Law**

$$n_1 \bullet \sin \Theta_1 = n_2 \bullet \sin \Theta_2$$

in which n_1 & n_2 are the refractive indices of the initial and secondary materials respectively, and Θ_1 and Θ_2 are the incident and transmitted angles.

Snell's Law says that refraction (bending) of light cannot take place when the angle of incidence grows too large (as when light travels from a material with a high refractive index to one with a low refractive index). If the angle of incidence exceeds

a certain critical value (in which the product of n_1 & the sine of the angle equals or exceeds one.) light cannot exit.
If light cannot exit the material, it is reflected. The angle that is reflected is equal to the angle of incidence. The phenomenon just described is called Total Internal Reflection & it is what keeps light inside an optical fiber.

Types of Optical Fiber
The simplest fiber optic cable consists of two concentric layers of transparent materials. The inner portion (the core) transports the light. The outer covering (the cladding) must have a lower refractive index than the core, so the two are made of different materials. The cable used in this kit also has a jacket to protect the optical properties of the core and cladding.

Optical fiber is generally made from either plastic or glass. This kit uses plastic fiber, which is very easy to terminate, & does not require special tools. Plastic fiber is generally limited to uses involving distance of less than 100 meters because of high attenuation (higher loss).

Glass fiber on other hand has very low attenuation (low loss) & hard to cut, requires special end connections & is more expensive.

The core of fiber is typically made of silica doped with impurities, which increases the refractive index relative to pure silica. The cladding, which surrounds the fiber core & creates an optical interface, is typically made from pure silica. The outer buffer coating is a plastic cover.
The first type of fiber optic cable put to use was called Step Index. In this design the cladding has a different index of refraction from the core. The light bounces off the sides & is reflected back into the fiber core. The problem with this design is that the reflected light must travel a slightly longer distances than that which travels down the center of the fiber, thus limiting the max. transmission rate. This design was improved with the use of Graded Index Fiber. In this design, the index of refraction decreases in proportion to the distance away from the center of the fiber core. The light moves more quickly in the outer portion, thus compensating for the additional distance. The change in index also has the effect of 'bending'. The light reflects back towards the core. This change increases the transmission capacity. In the newest Single Mode design, the diameter of the fiber core is made so small that all the light travels in a straight line. Even the latest fiber optic facility in use today uses less than 5% of the max. theoretical capacity of a Single Mode Fiber.

Single Mode V/S Multimode
There are essentially two different types of fiber optic transmission schemes in use. Multimode fiber optic systems are typically used in short haul applications such as LANS & FDDI. Recent growth in multimode installations has been driven by LAN/FDDI applications. The term Multimode means that the diameter of the fiber optic core is large enough to propagate more than one mode (electromagnetic wave). Because of the multimode modes, the pulse that is transmitted down the fiber tends

to become stretched over distance. This is referred to as Modal Dispersion & has the effect of reducing available bandwidth.

FIG. 2: Modal dispersion of light

Typical wavelengths used in multimode applications are 850 nm & 1300 nm. There are other wavelengths used (e.g. 660 nm for plastic fiber, 820/870 nm) in specialized applications such as the military. The vast majority of multimode applications are 850 or 1300 nm wavelengths. Most multimode cables in use today are graded index in type with a cladding diameter of 125 microns & fiber core diameter of 62.5 or 50 microns (Typically referred to as 62.5/125 or 50/125 fiber) Single Mode Systems are generally used in long haul applications (10’s or 100’s of kilometers) & local loop applications in both the carrier & CATV markets. Some large utilities such as electrical power companies also utilize single mode fiber optic backbones. As the name implies single mode fiber is designed to propagate only one mode of light. It is therefore not affected by modal dispersion & has higher bandwidth capacity. It is considered to be higher quality transmission system. Wavelengths used in single mode applications are 1300 & 1550 nm. The majority of today’s installed systems uses 1300 nm wavelength technology & is designed for higher speed applications. These systems are more sensitive to back reflections from connectors & sharp cable bends. They often require additional tests, such as Optical Return Loss, to be run. Most single mode cables in use today are step index in type with cladding diameter of 125 micron & fiber core diameter of 9 micron (Typically referred to as 9/125 fiber).

Advantages Of Fiber Optics Over Conventional Copper Cables

- Much grater bandwidth (Several orders of magnitude are theoretically possible).
- Immunity to electrical disturbances, ground loops, cross talks etc. in addition, no EMI radiation is generated)
- Much lighter. A 3-inch bundle of 900 twisted copper pairs can carry about 21,000 channels of traffic & weighs about 25,000 pounds/mile. A ½ inch fiber link with 12 fiber strands can carry about 3,00,000 channels & weighs about 200 pounds/mile. In addition less duct space is required to make the cable within building.
- Better in hostile environment, not affected as much by temperature, water etc.
- Lower transmission loss. Typical loss on a fiber link is 0.2 dB/Km. On a copper based facility, one can usually expect at least 5 db/km. For longer links, fewer repeaters are required.

- Better security as it is not possible to simply bridge onto the facility & monitor the traffic.

With all these properties of optical fiber, it is becoming very important in sensing field also. Most of the properties of optical fiber are useful for sensors. Further optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

ABOUT FCL-03

The purpose of ADVANCE FIBER OPTIC COMMUNICATION LAB is to provide you hands on experience on the various Fiber Optic & Digital Communications Techniques. In preparing this manual, we assumed you have a basic grasp of digital & transistor circuits. The experiments that can be performed with this Lab have been designed in a ladder fashion.
You have to start from the first step & then go for the next. Experiments start with basic principles of fiber optics & then go on to more sophisticated techniques. Every advanced experiment uses concept, procedures, and results of the previous experiments. So you are advised not to skip in between experiments. In this manual before starting the procedure of every experiment, objective of the experiment & basic theory related to that is given. You are advised to read it before starting the experiment because it will help you to understand the experiment very well. Also wherever necessary refer the datasheets of various components provided on the backside of the manual.

Before starting the experiments check following things in the kit.

- Plastic fiber of 1Meter length
- 1 No. Power Supply
- 2 Nos. Serial cables
- Link set containing Patch chords, Jumpers & crocodile clips wires
- Experimental and Circuit Description Manuals

Optical Fiber Preparation Instructions

The fibers provided with this Lab are multimode, step index, 1000-micron plastic fiber & are factory prepared for use. They are not require to prepared again even after several operations, if handled carefully. In case the fiber is broken or damaged you have to prepare each end of the optical fiber cable very carefully so it transmits light effectively.

- Cut off the ends of the cable with a single-edge razor blade or sharp knife. Try to obtain a precise 90-degree angle.
- Wet the polishing paper with water or light oil & place it on flat, firm surface. Hold the optical fiber upright, at right angle to the paper & polish the fiber tip with a gentle “figure8” motion. You may get the best result by supporting the upright fiber against some flat object such as a portion of the printed circuit board.

- Using an 18-gauge wire stripper, remove 3 mm of the jacket from the end of the fiber. Be careful not to nick the buffer in the process or you will be losing light later.

CIRCUIT DESCRIPTION

1.1. Analog Buffer

This is a unity gain amplifier built around op-amp IC LF357 (U8 & U11). Output voltage is limited to 5.1Vpp using zener diode at the output post of buffer.

1.2. Clock Generator

Refer to the circuit diagram as shown later in this manual sheet **2** of **9** of **FCL-03**. The 2.048MHz crystal oscillators generate a 2MHz clock. This clock is applied to dual counter U14 (IC 74HCT393). Counter will generate 256KHz clock signal.

1.3. Digital Buffer

U18 (IC 74HCT04) is used as TTL Buffer.

1.4. Fiber Optics Transmitter

The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver.

Fiber optic transmitters are typically composed of a buffer, driver & optical source. The buffer section provides both an electrical connection & isolation between the transmitter & the electrical system supplying the data. The driver section provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally the optical source (LED) converts the electrical current to light energy with the same pattern.

FCL-03 kit supply SFH756V LED transmitter. The LED SFH756V operates on the visible light spectrum. Its optical output is centered at visible red light wavelength of 660nm. This LED is coupled to the transistor driver in common emitter mode. In the absence of input signal half of the supply voltage appears at the base of the transistor. This biases the transistor near midpoint within the active region for linear applications. Thus LED emits constant intensity of light at this time. When the signal is applied to transmitter it overrides the DC level at the base of the transistor, which causes the Q point of the transistor to oscillate about the midpoint. So the intensity of the LED varies about its previous constant value. This variation in the intensity has linear relation with the input electrical signal. Optical signal is then carried over by the optical fiber.

Jumper JP2 provides VCC signal to anode of SFH756V. Jumper JP3 is used to select either Digital or Analog driver stage. Intensity of signal can be varied using pot P3 in analog driver stage.

1.5. Fiber Optics Receiver1

At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter.

The function of the receiver is to convert the optical energy into electrical form, which is then conditioned to reproduce the transmitted electrical signal in its original form. The detector SFH250V used in the FCL-03 has a diode type output. The parameters usually considered in the case of detector are its responsivity at peak wavelength & response time. SFH250V has responsivity of about 4μA per 10 μW of incident optical energy at 950 nm and it has rise & fall time of 0.01 μ sec.

PIN photodiode is normally reverse biased. When optical signal falls on the diode, reverse current start to flow, thus diode acts as closed switch and in the absence of light intensity, it acts as an open switch. Since PIN diode usually has low responsively, a transimpedance amplifier is used to convert this reverse current into voltage. This voltage is then amplified with the help of another amplifier circuit. This voltage is the duplication of the transmitted electrical signal.

1.6. Fiber Optics Receiver2

At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter. Detector SFH551V is a phototransistor detector with TTL logic output. When light falls on window of this detector it gives TTL high output and TTL low output when light is absent.

1.7. Pulse Width Modulation

Pulse Width Modulation this technique of modulation controls the variation of duty cycle of the square wave (With some fundamental frequency) according to the input modulating signal. Here the amplitude variation of the modulating signal is reflected in ON period variation of square wave. Hence, it is also called as technique of V to T conversion.

The circuit is divided in to two stages. First one is the circuit built around Op-Amp U20 (IC LF 353). This is unity gain non-inverting amplifier designed as level shifter. When there is no input signal, the output of the amplifier simply follows the voltage at inverting input terminal, with inverted polarity. This is necessary because PWM chip requires only positive polarity signal. Now the input signal fed to non-inverting terminal over rides this shifted voltage.

This signal is now fed to second part of the circuit. This is specialized for regulating Pulse Width Modulation. This primarily consists of one fixed frequency oscillator whose frequency is decided by timing register Rt and capacitor Ct. The regulation is simply given by formula

$$\mathbf{F = 1.18 / (Rt \times Ct)}$$

Where F is in KHz, R is in KΩ and C is in μF. Also an error amplifier, comparator and flip-flop are important functioning parts internal to the IC. When no input signal is given to the error amplifier the output oscillates with the same frequency decided by the external R & C network. When input is given to the error amplifier, a linear charging of the Ct results in a linear voltage ramp, which is then fed to the comparator. This provides the linear control of the output pulse width. The amplifier output is compared with the linear charging voltage at Ct, resulting in the modulated pulse at the output of high gain comparator. Thus we get the PWM. For more details please refer to the SG 3524 data sheets.

1.8. Pulse Position Modulation

The position of the TTL pulse is changed on time scale according to the variation of input modulating signal amplitude. The circuit uses a single U24 (IC 74HC123) which is dual monostable multivibrator. The ON period of the output pulse from the monostable is decided by the external R and C network. Here, the active low reset input pin (3) is tied to VDD. Here trailing edge triggering is used. Output is taken from pin no. 13 of monostable-2 (Q).

Now Pulse Width Modulated signal is fed as input to this circuit. Please note that input-modulating signal must be converted into Pulse Width Modulated form before applying to Pulse Position Modulator. As the signal is PWM, naturally, according to the input signal, the pulse duration is changing and this change in pulse duration causes for the delay in triggering. The input is given to trailing edge trigger input of monoshot. So finally we get the pulses at the output, which are shifted on the time slot. This is nothing but pulse position modulation.

Thus Pulse Positions are directly proportional to the instantaneous values of modulating signal.

1.9. Pulse Width Demodulation

The input signal is Pulse Width Modulated, so the ON time of the signal is changing according to the modulating signal. In this demodulation technique the PWM signal is applied to IC U21 (IC LF353) which acts as an integrator from which the signal emerges whose amplitude at any time is proportional to the pulse width at that time. This Integrated output is then filtered to get original signal.

1.10. Pulse Position Demodulation

The pulse position modulated signal is ORed with pulse generated by the rising edge of pulse width modulated signal using U25 (IC 74LS32). The output of the OR gate is fed to CLK input of flip-flop U26 (IC 74LS74). Thus flip-flop acts as a bistable multivibrator giving out high output for the duration between rising edge of PWM signal & PPM signal. Since PPM corresponds to the end of PWM pulse output of flip-flop is exactly same as that of PWM signal. This signal is then demodulated using the same technique of Pulse Width demodulation.

1.11. FDM Transmitter

1.111. Frequency Modulator 1 & 2

In Frequency Modulation the Carrier frequency is varied in accordance with the amplitude of input signal. FM Modulation is generated by using IC U2 & U3 (IC ICL8038). The frequency of the waveform generator is a direct function of the DC voltage at pin no. 8 of IC U2 & U3 (IC ICL8038), (measured from V+). By altering this voltage, frequency modulation is performed. For small deviations (e.g. ± 10%) the modulating signal can be applied directly to pin no.8 of U2 & U3, merely providing DC decoupling with a capacitor as shown. The FM output obtained is fed to U5 & U6 (IC CA3140) which acts as a buffer.

1.112. Band Pass Filter 1

This filter is built around op-amp U27 (IC 741) with pass band frequency range of 8KHz to 12KHz and cut off frequency of 10KHz.

1.113. Band Pass Filter 2

This filter is built around op-amp U28 (IC 741) with pass band frequency range of 18KHz to 22KHz and cut off frequency of 20KHz.

1.114. Adder

Adder is formed using U32 (IC 4053). In this integrated chip will do summation of two signals applied to it. If two signals with different frequency components are applied to this chip we will get FDM output.

1.12. Filter 1 & 2

Refer to the circuit diagram as shown later in this manual sheet **7** & **9** of **9** of **FCL-03**. Fourth order low pass butterworth Filter is formed by U33 & U34 (IC KF347).

1.13. FDM Receiver

1.131. Band Pass Filter 3

This filter is built around op-amp U30 & U31 (IC 741) with pass band frequency range of 8KHz to 12KHz and cut off frequency of 10KHz.

1.132. Band Pass Filter 4

This filter is built around op-amp U29 (IC 741) with pass band frequency range of 18 KHz to 22 KHz and cut off frequency of 20KHz.

1.14. Frequency Demodulator 1 & 2

Frequency Demodulation is carried out using Phase Lock Loop Technique (PLL) with the help of U35 & U36 (IC 74HC4046). FM signal is applied to pin no.14 & RC combination at pin no.11 decides the Locking frequency range of the PLL. Demodulated signal is further filtered out using filter formed by U33 & U34 (IC KF347)

1.15. RS 232 Interface

U37 (IC MAX232C) is standard RS232 interface IC is used. D type 9 pin connectors are used for Interfacing PC’s.

1.16. Switch Faults

Eight switch faults are provided to insert logical faults in various sections of the circuit.

FG-02 FUNCTION GENERATOR

2.1 Variable Frequency Generator

The integrated circuit U1 (ICL8038) generates sine (pin 2) and triangular waveform (pin 3). The capacitors which can be connected to pin 10 enable to select the range (10Hz, 100Hz, 1KHz, 10KHz or 100 KHz max). The frequency is varied with the pot **P1 FREQ.** (P3 limits the inferior frequency). The trimmers P5 and P6 adjust the symmetry and the form of the sine signal.

With the Jumpers JP1 & JP2 the sine and the triangular signals are applied to the amplifier U2 (LF353) and then to the output. The pot **P2** varies the signal amplitude

With Jumper JP3 the square signal can be applied to the output amplifier. Frequency range can be selected with help of switch SW1 and can be observed at **OUT** post.

From the square wave signal TTL signal is also generated using U4 (IC 74HCT14)

2.2 Clock Generator

The timer U4 (IC NE555) is used to generate the clock waveform of 32 KHz.

2.3 Fixed Frequency Sine Wave Generator

This basically provides amplitude variable (0 - 5 Vpp) synchronized sine wave, of 2KHz.

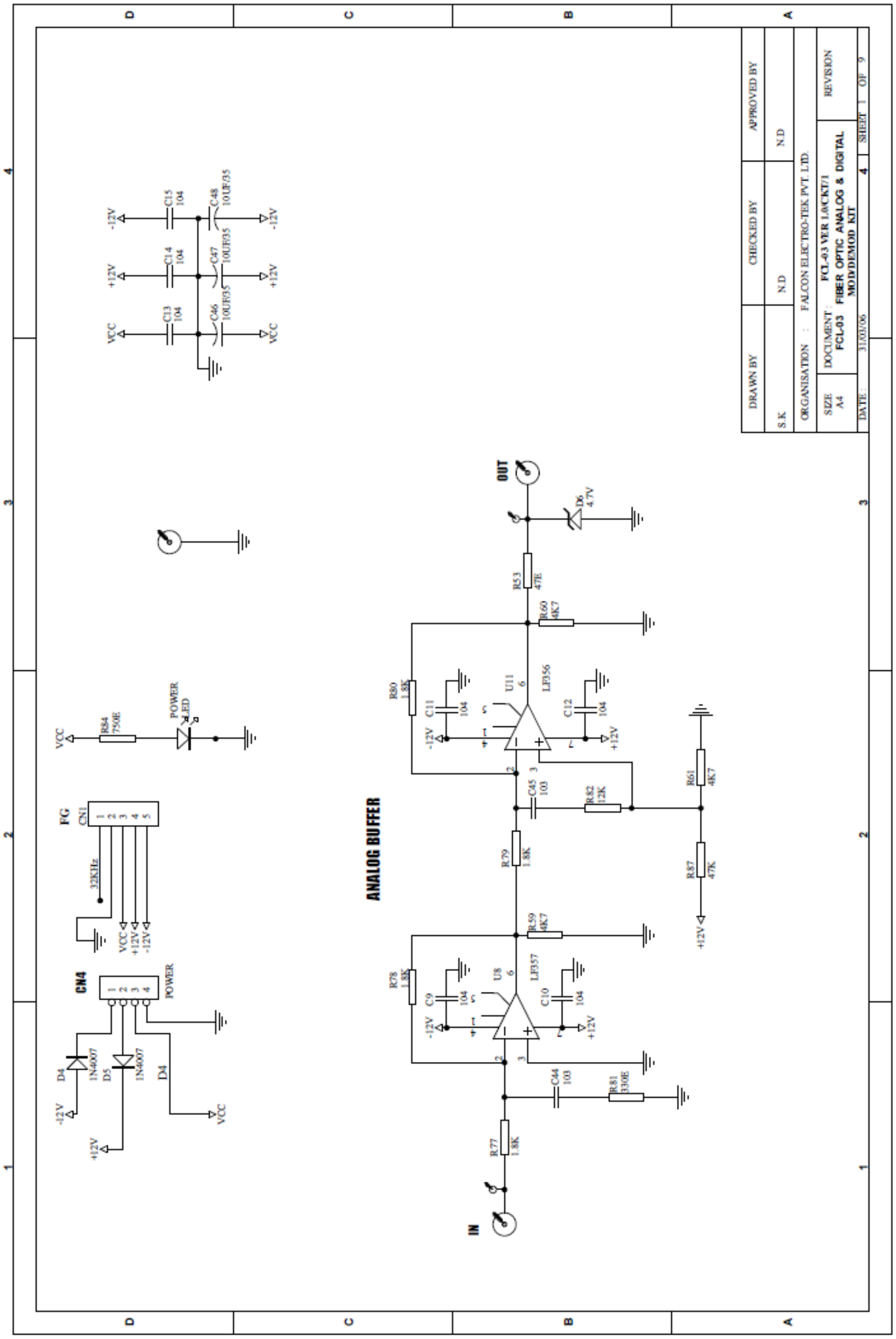
The 32KHz frequency generated in clock and timing section is fed to serial to parallel shift register U6 (IC 74HC164), which generates the sine wave, by serial shift operations. The serial to parallel shift registers (IC 74HC164) have resistive ladder network at the output. For 16 shifts of the register, one stare case sine wave is produced. So if a 32KHz clock is fed to the shift register, 2Khz sine wave is generated. The active filter U7 (IC KF347) at the output suppresses the ripple and also takes care of the impedance matching. This gives smooth sine waves. The potentiometers P7 is used to vary amplitudes of 2KHz sine wave.

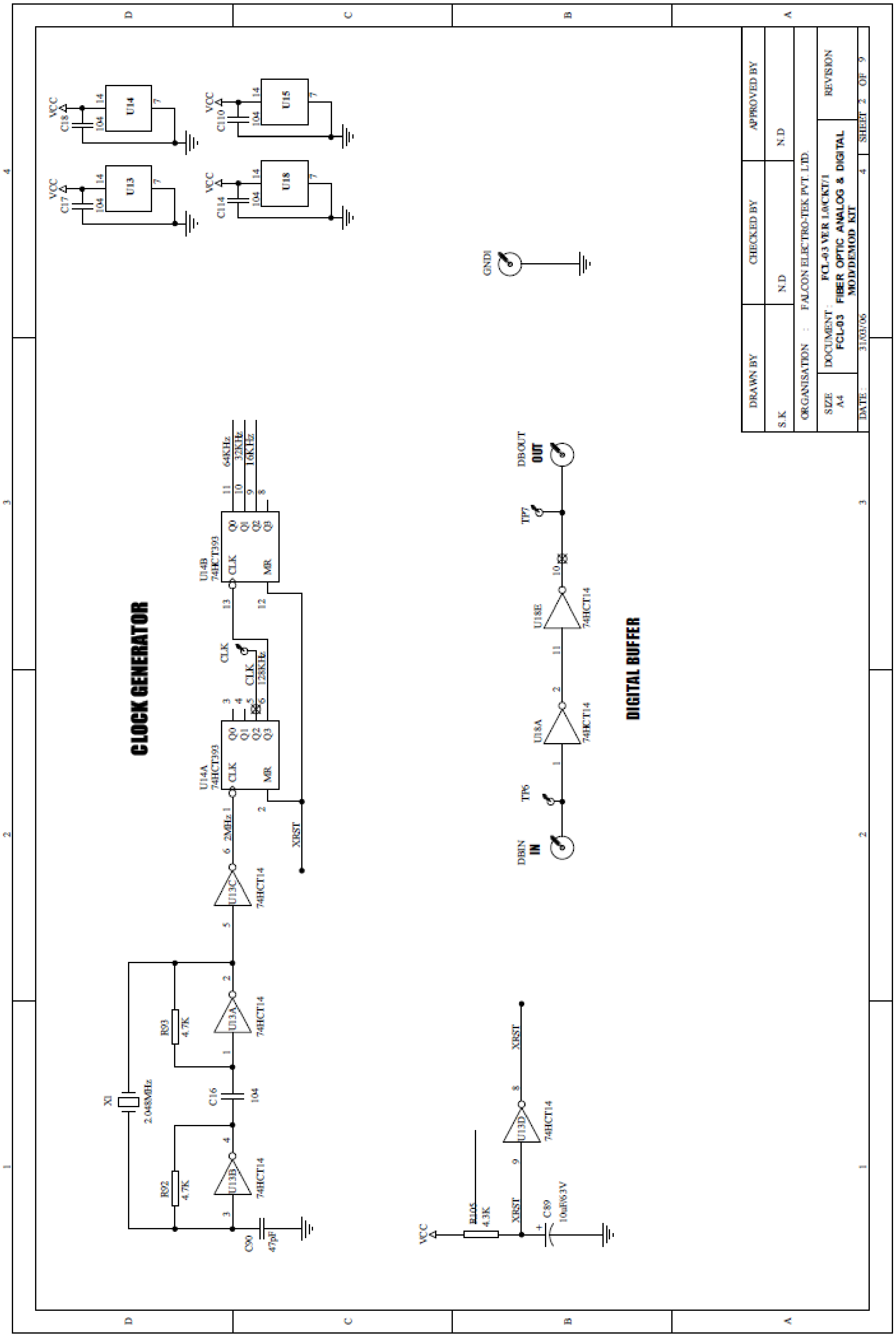
SWITCH FAULTS

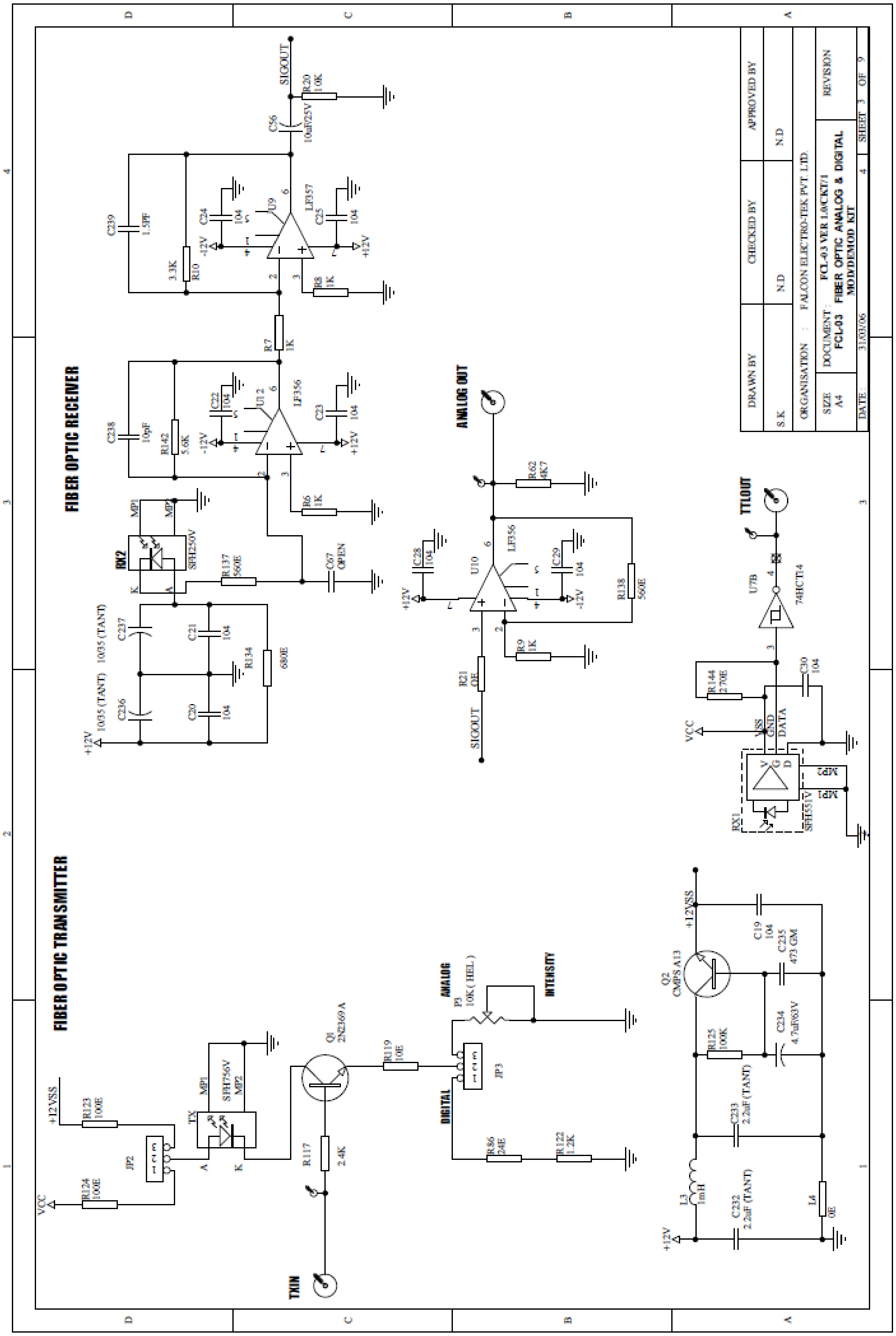
Switch fault section on FCL-03 is provided with 8 switches. These switches can be Used to simulate fault conditions in various parts of the circuit. In order to keep FCL-03 kit fully operational, all switches should be set to OFF position before use. Switch faults are introduced one at a time.

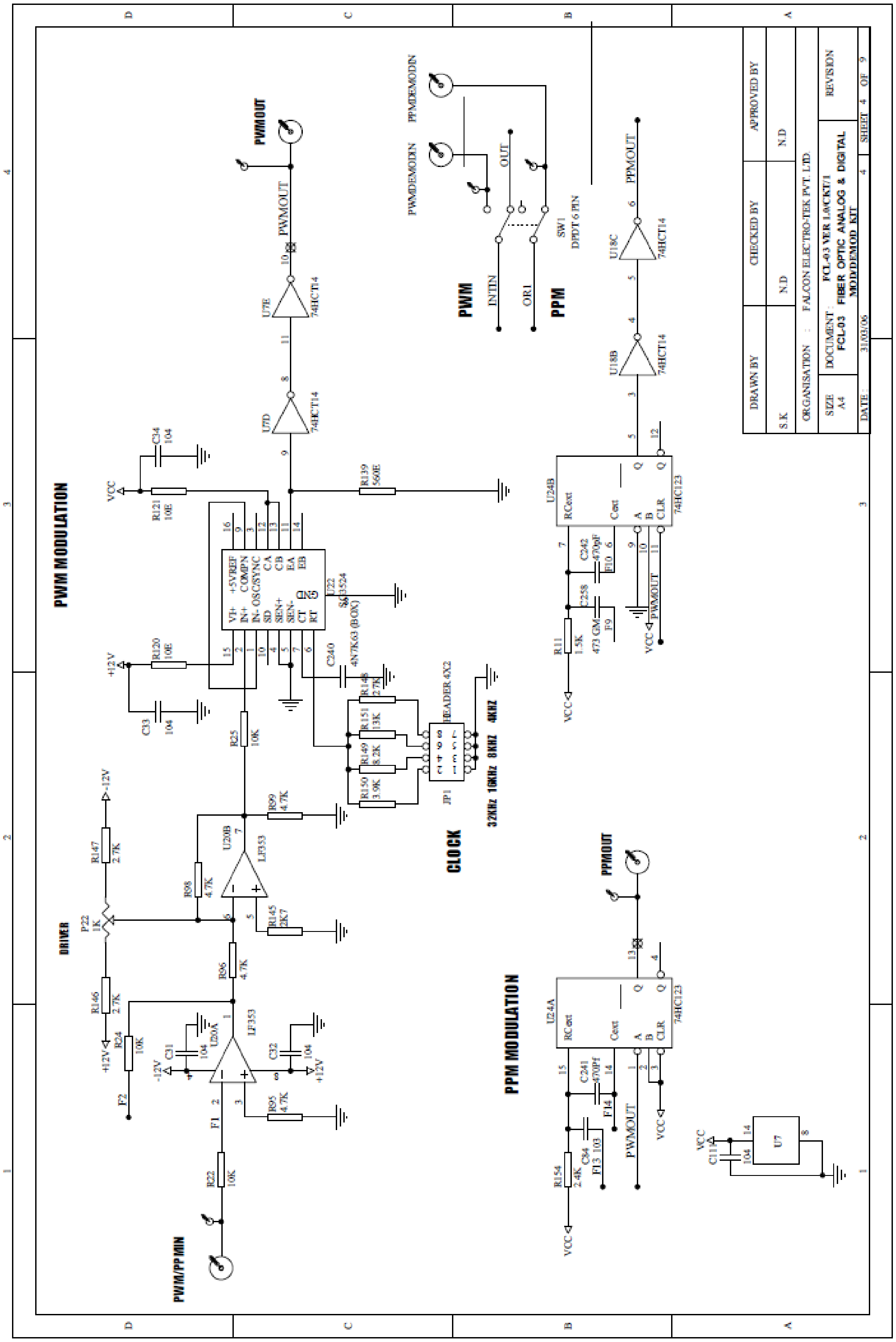
- Put switch **1 (SF1)** in Switch Fault section to ON position. This will open Resistor R24 from the feedback loop of PWM modulator driver stage. U20 will operate in open loop mode & o/p gets distorted.
- Put switch **2 (SF1)** in Switch Fault section to ON position. This will open Capacitor C51 from second stage of Band Pass Filter 3 & center frequency of BPF no.3 gets distorted.
- Put switch **3 (SF1)** in Switch Fault section to ON position. This will open Resistor R47 in series with Potentiometer P14 from the FM demodulator 2. This will result in change of free running freq.
- Put switch **4 (SF1)** in Switch Fault section to ON position. This will open Capacitor C223 & C218 from first stage of Filter 2. Filter o/p gets distorted.
- Put switch **5 (SF2)** in Switch Fault section to ON position. This will short Capacitor C258 & C242 at PPM demodulator first stage & duty cycle of waveform will change.
- Put switch **6 (SF2)** in Switch Fault section to ON position. This will short Capacitor C85 & C244 at PPM demodulator final stage. This will cause distortion in integrator response.
- Put switch **7 (SF2)** in Switch Fault section to ON position. This will short Capacitor C84 & C241 at PPM modulator stage. This will cause change in width of the pulse.
- Put switch **8 (SF2)** in Switch Fault section to ON position. This will short Resistor R135 & R168 from the Band Pass Filter 1. This will result in distorted output.

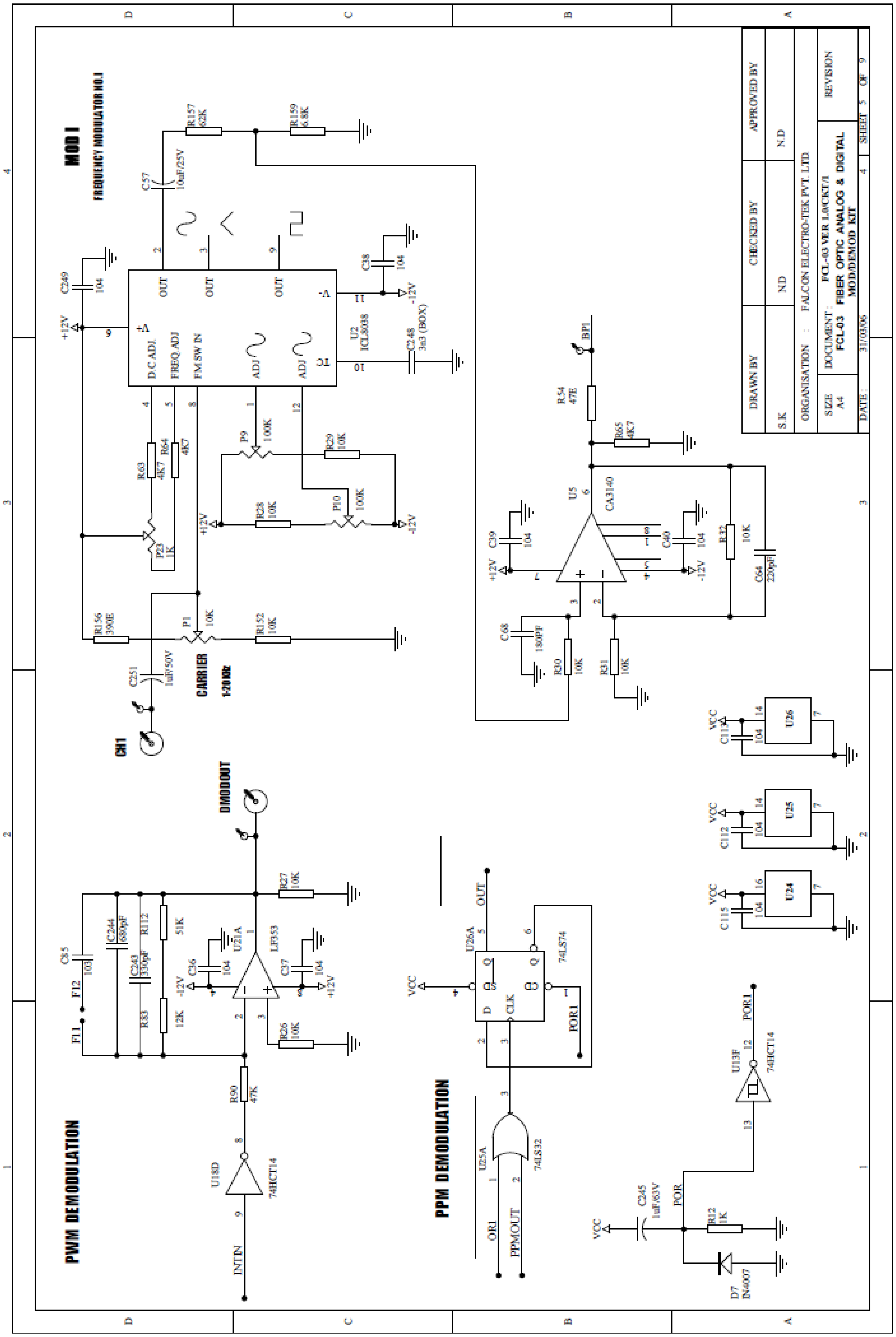
CIRCUIT DIAGRAM
FCL-03

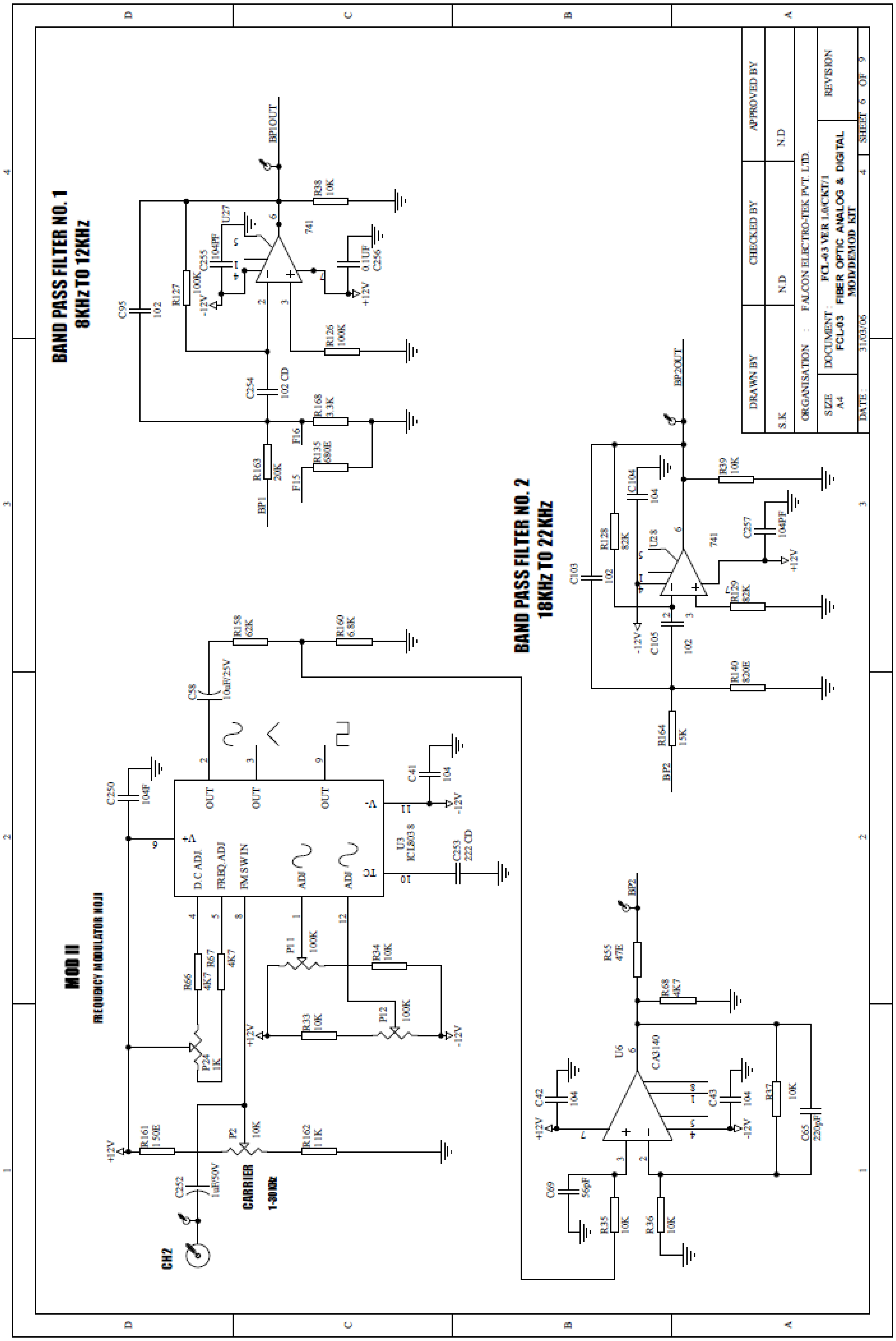


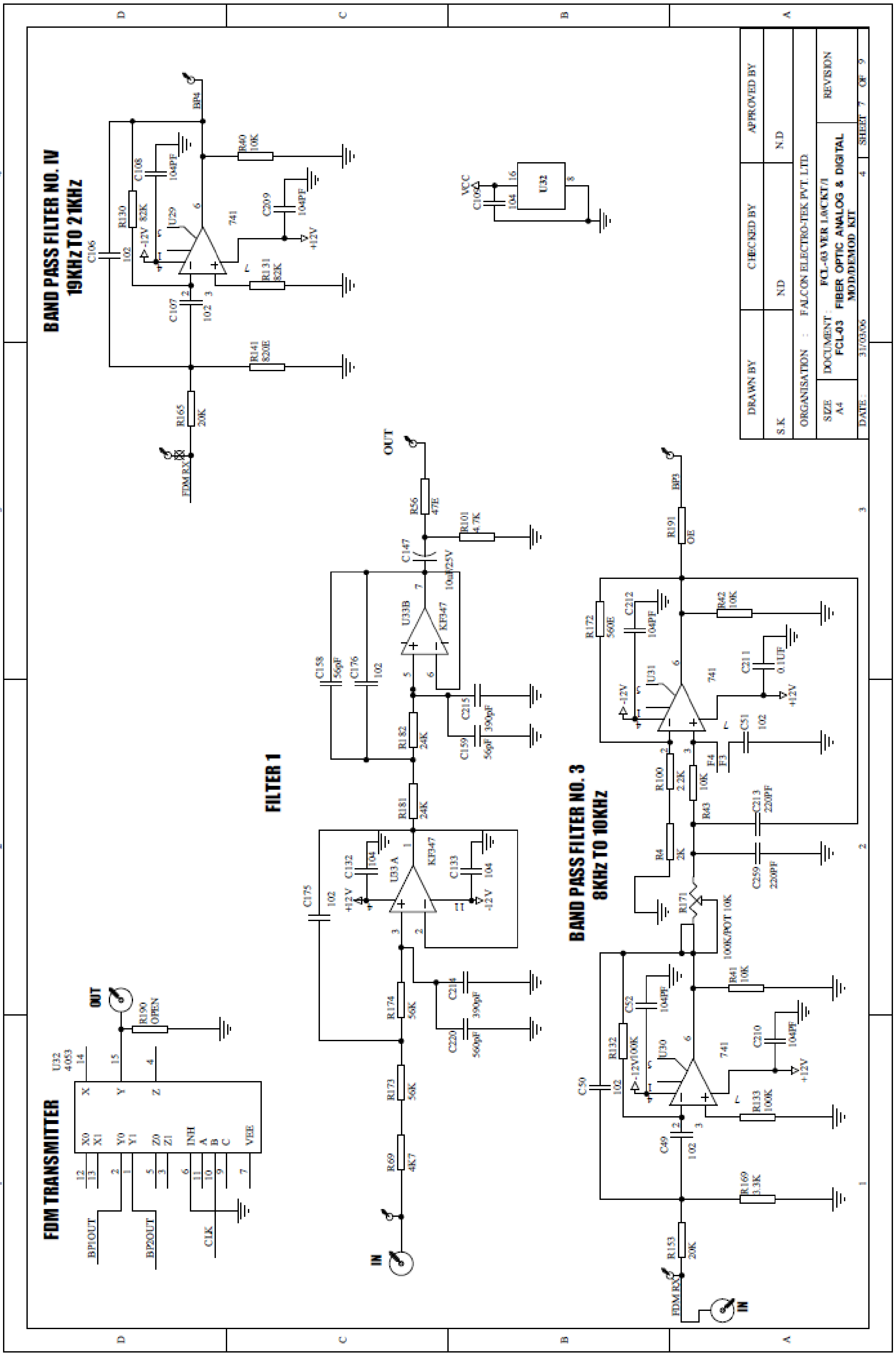

CLOCK GENERATOR
DIGITAL BUFFER
FALCON ELECTRO-TEK PVT. LTD.
FIBER OPTIC ANALOG & DIGITAL MOD/DEMOD KIT

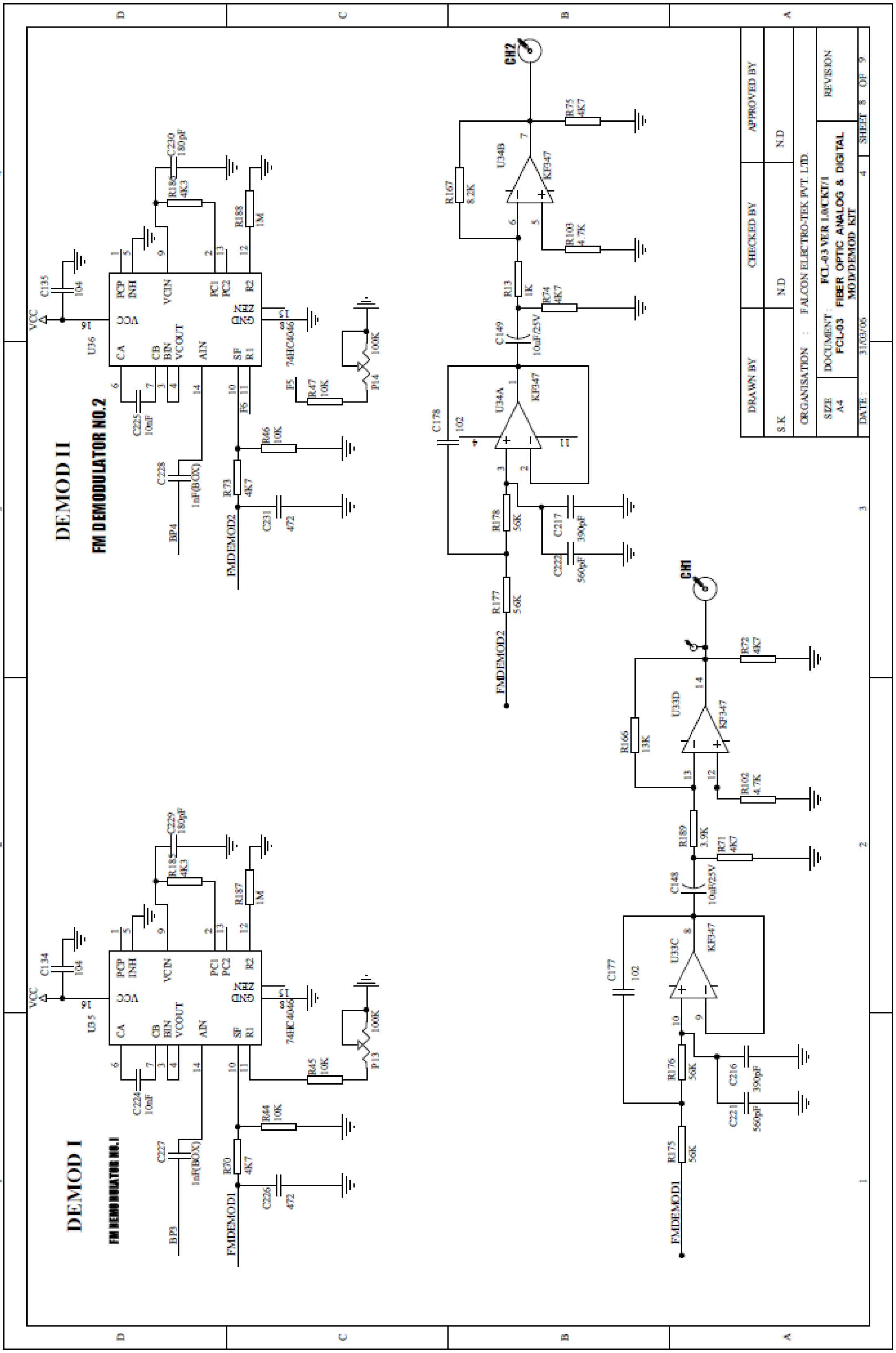

FIBER OPTIC RECEIVER
FIBER OPTIC TRANSMITTER
ANALOG OUT
TTLOUT
TXIN
SIGOUT
ANALOG
DIGITAL
INTENSITY
DRAWN BY
CHECKED BY
APPROVED BY
S.K
N.D
N.D
ORGANISATION : FALCON ELECTRO-TEK PVT LTD
DOCUMENT : FCL-03 VER 1.0/CKT/1
FCL-03 FIBER OPTIC ANALOG & DIGITAL MOD/DEMOD KIT
SIZE A4
REVISION
DATE 31/03/06
SHEET 3 OF 9


PWM MODULATION
PPM MODULATION
CLOCK
DRIVER
PWMOUT
PPMOUT
PWM/PPMIN
PWMDEMODN
PPMDEMODN
PWM
PPM
SW1
DPDT 6 PIN
HEADER 4X2
32KHz 16KHz 8KHz 4KHz
DRAWN BY
CHECKED BY
APPROVED BY
S.K
N.D
N.D
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD.
REVISION
FCL-03 VER 1.0/CKT/1
DOCUMENT : FCL-03
FIBER OPTIC ANALOG & DIGITAL MOD/DEMOD KIT
SIZE A4
DATE : 31/03/06
SHEET 4 OF 9

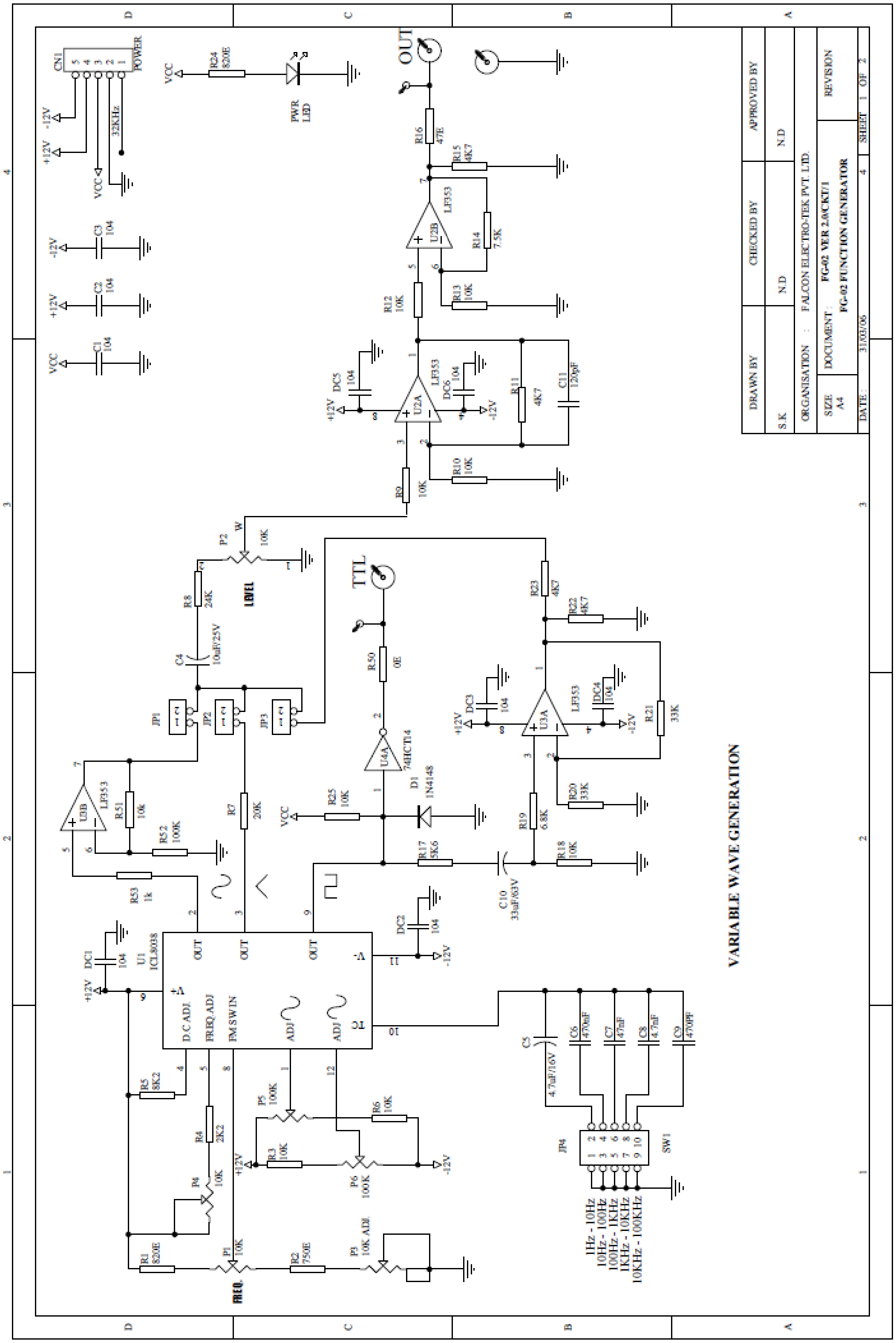

MOD I
FREQUENCY MODULATOR NO.1
CARRIER
1-20KHz
CH1
DMODOUT
PWM DEMODULATION
PPM DEMODULATION
ICL8038
CA3140
LF353
74HCT14
74LS74
74LS32
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD
DOCUMENT : FCL-03 VER 1.0/CKT/1
FCL-03 FIBER OPTIC ANALOG & DIGITAL MOD/DEMOD KIT
DRAWN BY S.K
CHECKED BY N.D
APPROVED BY N.D
SIZE A4
DATE 31/03/06
SHEET 5 OF 9
REVISION

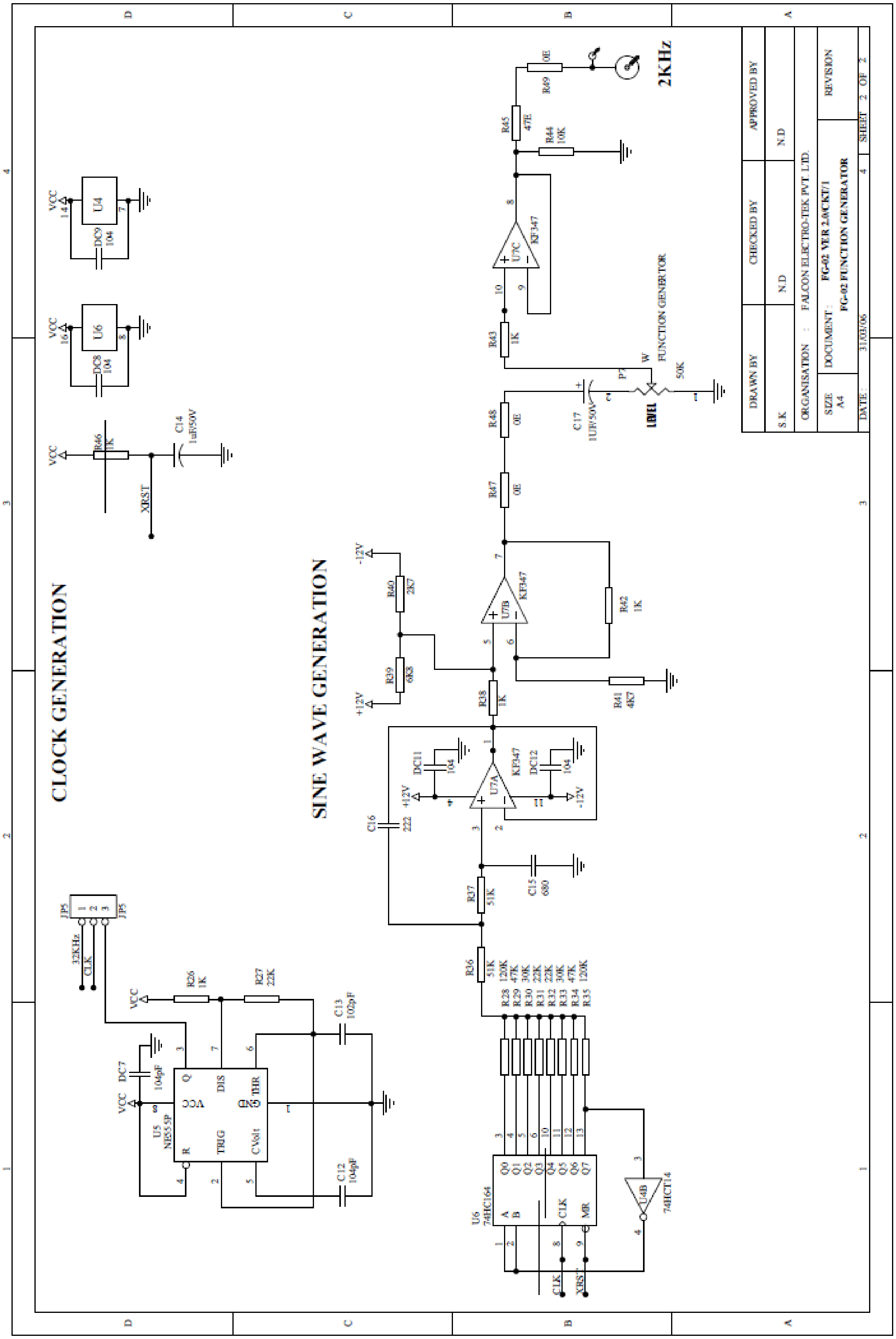

BAND PASS FILTER NO. 1
8KHz TO 12KHz
BAND PASS FILTER NO. 2
18KHz TO 22KHz
MOD II
CARRIER
CH2
BP1OUT
BP2OUT
FALCON ELECTRO-TEK PVT. LTD.
FCL-03 FIBER OPTIC ANALOG & DIGITAL MOD/DEMOD KIT
31/03/06


BAND PASS FILTER NO. IV
19KHz TO 21KHz
FILTER 1
FDM TRANSMITTER
BAND PASS FILTER NO. 3
8KHz TO 10KHz


DEMOD I
FM DEMODULATOR NO.1
DEMOD II
FM DEMODULATOR NO.2
74HC4046
KF347
CH1
CH2
FMDEMOD1
FMDEMOD2
DRAWN BY
CHECKED BY
APPROVED BY
S.K
N.D
N.D
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD.
DOCUMENT : FCL-03 VER 1.0/CKT/1
FCL-03 FIBER OPTIC ANALOG & DIGITAL MOD/DEMOD KIT
SIZE A4
REVISION
DATE : 31/03/06
SHEET 8 OF 9

FG-02


VARIABLE WAVE GENERATION
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD.
DOCUMENT : FG-02 VER 2.0/CKT/1
FG-02 FUNCTION GENERATOR
DRAWN BY S.K
CHECKED BY N.D
APPROVED BY N.D
SIZE A4
DATE : 31/03/06
SHEET 1 OF 2
OUT
TTL
ICL8038
LF353
74HCT14
LEVEL
FREQ.
POWER


CLOCK GENERATION
SINE WAVE GENERATION
2KHz
FUNCTION GENERATOR
LEVEL
DRAWN BY
CHECKED BY
APPROVED BY
S.K
N.D
N.D
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD.
DOCUMENT : FG-02 VER 2.0/CKT/1
FG-02 FUNCTION GENERATOR
SIZE A4
REVISION
DATE : 31/03/06
SHEET 2 OF 2

EXPERIMENT NO.1

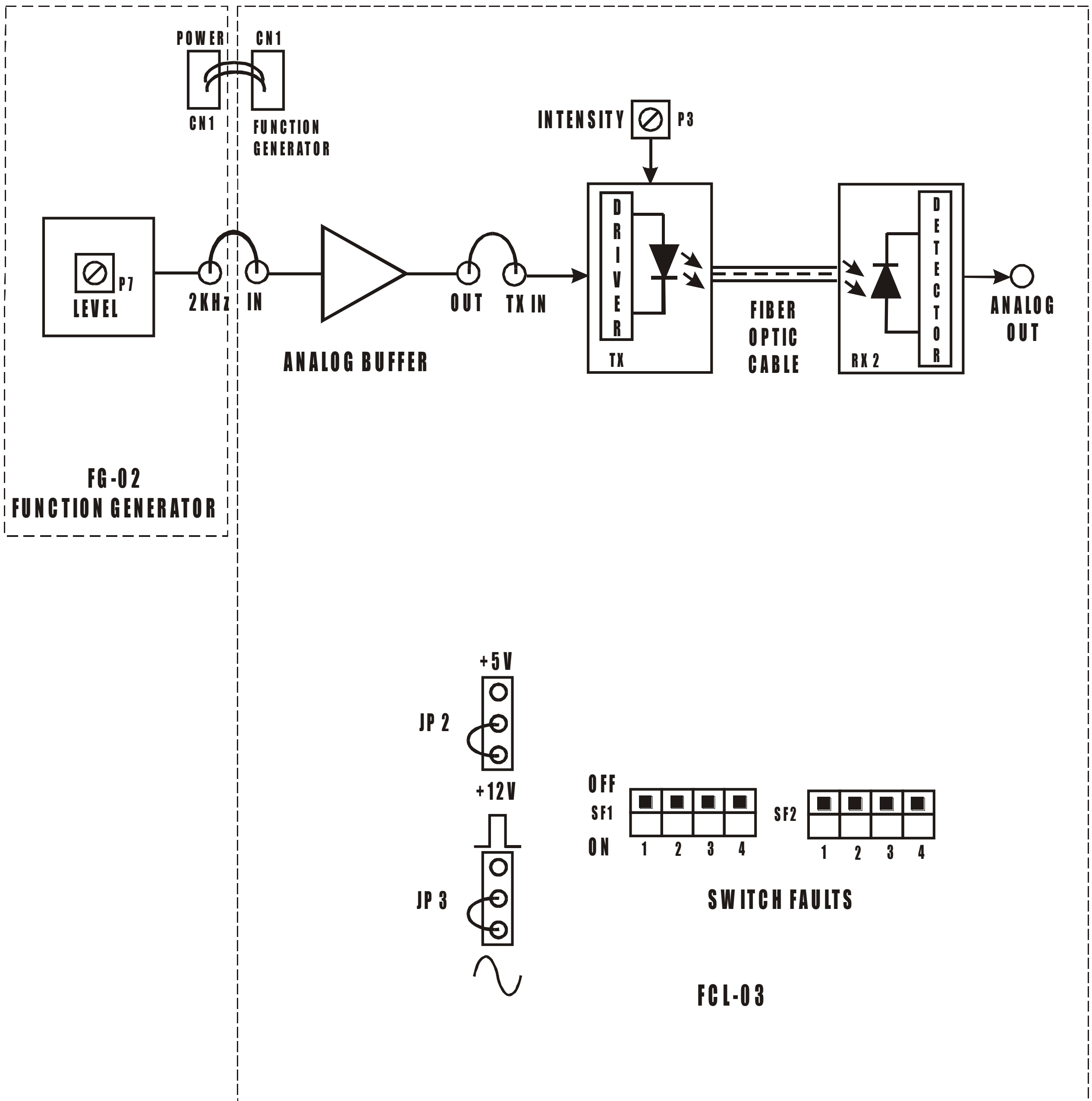


FIG. 1.1 BLOCK DIAGRAM FOR FIBER OPTIC ANALOG LINK

EXPERIMENT NO: 1

NAME

Setting Up a Fiber Optic Analog Link

OBJECTIVE

The objective of this experiment is to study a 660 nm Fiber Optic Analog Link. In this experiment, you will study the relationship between the input signal & received signal.

THEORY

Fiber Optic Links can be used for transmission of digital as well as analog signals. Basically, a fiber optic link contains three main elements, a transmitter, an optical fiber & a receiver. The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter.

EQUIPMENTS

- FCL-03
- FG-02 with power cable
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position

PROCEDURE

- Make connections as shown in fig.1.1. Connect the power supply cables with proper polarity to FCL-03 Kit. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-02 to FCL-03 using power cable.
- Switch on the power supply.

- Keep the jumpers **JP2** & **JP3** on FCL-03 as shown in fig.1.1.
- Connect the **2KHz, 2Vpp** Signal from FG-02 as a constant signal to the **IN** post of Analog Buffer on FCL-03.
- Connect the output of Analog Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Now rotate the Optical Power Control pot **P3** in FCL-03 in anticlockwise direction. This ensures minimum current flow through LED.
- Slightly unscrew the cap of **RX2** Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Observe the output signal from the detector at **ANALOG OUT** post on Oscilloscope by adjusting Optical Power Control Pot **P3** in clockwise direction and you should get the reproduction of the original transmitted signal.
- To measure the analog bandwidth of the link, keep the same connections and vary the frequency of the input signal from 100 Hz onwards. Measure the amplitude of the received signal for each frequency reading.
- Plot a graph of Gain/Frequency. Measure the frequency range for which the response if flat.

EXPERIMENT NO.2

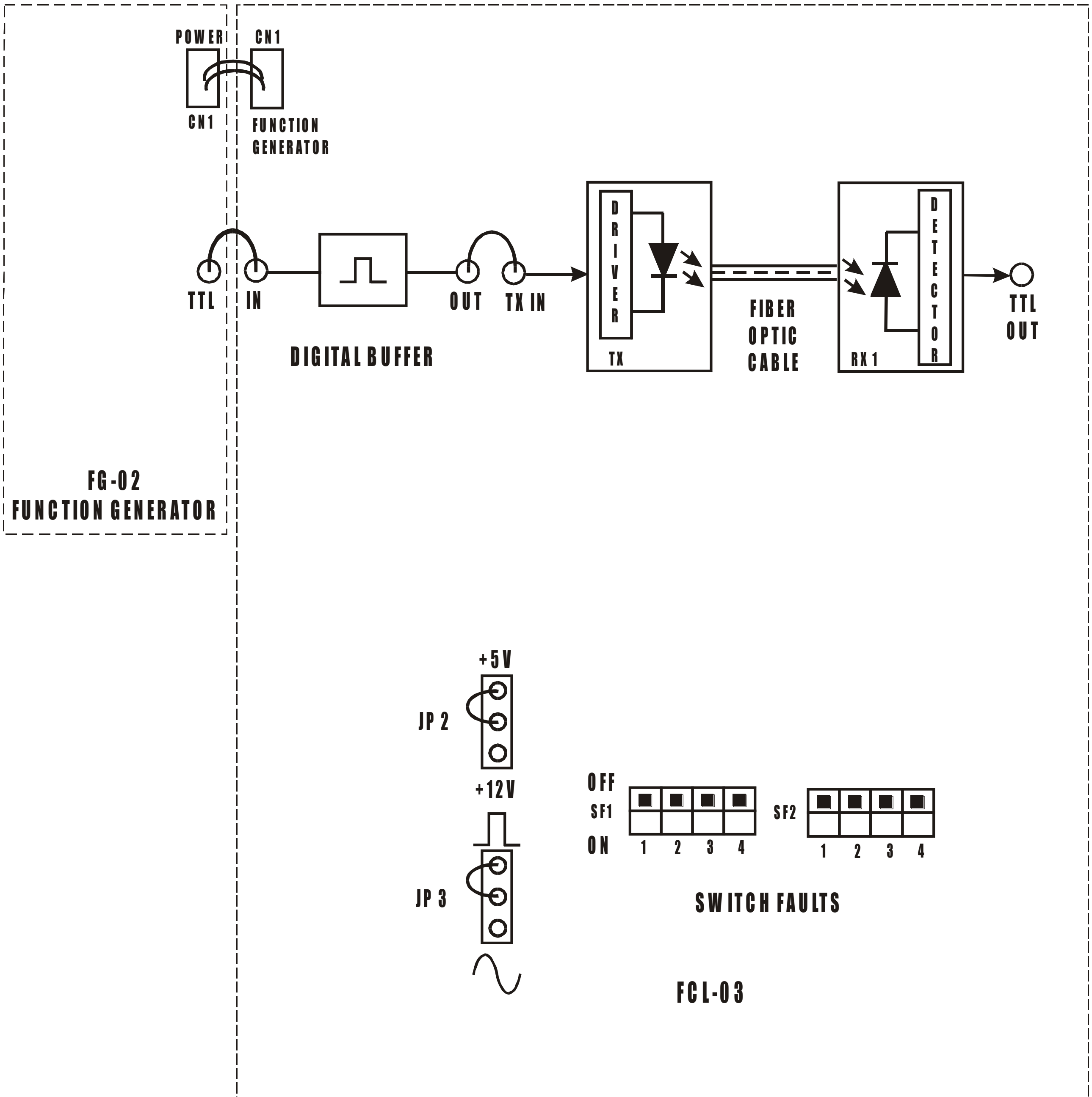


FIG. 2.1 BLOCK DIAGRAM FOR FIBER OPTIC DIGITAL LINK

EXPERIMENT NO.2

NAME

Setting Up Fiber Optic Digital Link

OBJECTIVE

The objective of this experiment is to study 660 nm fiber optic digital link. Here you will study how digital signal can be transmitted over fiber cable and reproduce at the receiver end.

THEORY

In the experiment no. 1, we have seen how analog signal can be transmitted and received using LED, fiber and detector. The same LED can be configured for the digital applications to transmit binary data over fiber. Thus basic elements of the link remain same even for digital applications.

Transmitter
LED, digital, DC coupled transmitters are one of the most popular variety due to their ease of fabrication. We have used a standard TTL gate to drive a NPN transistor, which modulates the LED SFH756V (660nm) source (Turns it ON and OFF).

Receiver
There are various methods to configure detectors to extract digital data. Usually detectors are of linear nature. We have used a photodetector having TTL type output. Usually it consists of PIN photodiode, transimpedance amplifier and level shifter.

EQUIPMENTS

- FCL-03
- FG-02 with power cable
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.2.1. Connect the power supply cables with proper polarity to FCL-03 Kit. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-02 to FCL-03 using power cable.
- Switch on the power supply.
- Keep the jumpers **JP2** & **JP3** on FCL-03 as shown in fig.2.1.
- Connect the **TTL** Signal from FG-02 as a constant signal to the **IN** post of Digital Buffer on FCL-03.
- Connect the output of Digital Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of **RX1** Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Observe the output signal from the detector at **TTL OUT** post on Oscilloscope you should get the reproduction of the original transmitted signal.
- To measure the digital bandwidth of the link, keep the same connections and vary the frequency of the input signal from 100 Hz onwards. Observe the variation in duty cycle of the received signal for each frequency reading and determine the maximum bit rate that can be transmitted on the digital link.

EXPERIMENT NO.3

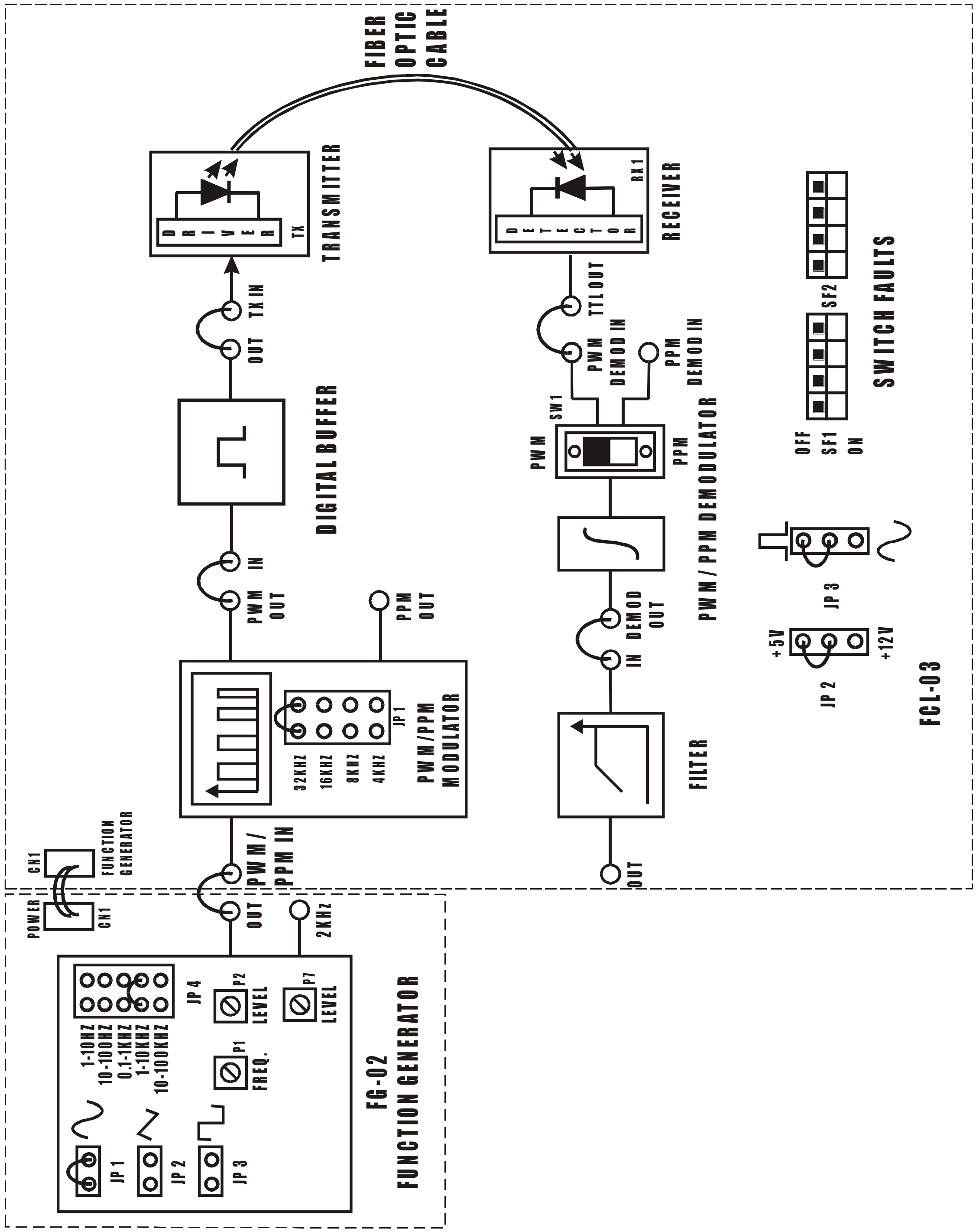


FIG. 3.1 BLOCK DIAGRAM FOR PULSE WIDTH MODULATION AND DEMODULATION

EXPERIMENT NO.3

NAME

Study of Pulse Width Modulation And Demodulation

OBJECTIVE

The objective of this experiment is to study the circuit action of Pulse Width Modulation and Demodulation over Fiber Optic Digital Link.

THEORY

Pulse Width Modulation
This technique of modulation controls the variation of duty cycle of the square wave (With some fundamental frequency) according to the input modulating signal. Here the amplitude variation of the modulating signal is reflected in to ON period variation of square wave. Hence, it is also called as technique of V to T conversion.

Pulse Width Demodulation
The input signal is pulse width modulated, so the ON time of the signal is changing according to the modulating signal. In this demodulation technique, the ON time of PWM signal is changing according to the modulating signal. In this demodulation technique, the PWM signal is applied to an Integrator, whose output is then Filtered to obtain original signal. Thus at the output we get the original modulating signal extracted from PWM wave. Fiber optic transmitter (SFH756v) & receiver (SFH551v) are used to transmit and receive PWM signal respectively.

EQUIPMENTS

- FCL-03
- FG-02 with power cable
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.3.1. Connect the power supply cables with proper polarity to FCL-03 Kit. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-02 to FCL-03 using power cable.
- Switch on the power supply.
- Keep the jumpers **JP1**, **JP2** & **JP3** on FCL-03 as shown in fig.3.1.
- Keep the function generator **JP1** shorted & **JP2**, **JP3** open.
- Keep the function generator **JP4** in **1-10KHz** position.
- Connect the **OUT** Signal from FG-02 to the **PWM/PPM IN** post of PWM/PPM Modulator on FCL-03 and keep the signal frequency at **1KHz** & Amplitude at **1Vpp**.
- Connect the **PWM OUT** Signal from PWM/PPM Modulator to the **IN** post of Digital Buffer on FCL-03.
- Connect the output of Digital Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of **RX1** Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect the output of detector post **TTL OUT** to post **PWM DEMOD IN** of PWM/PPM Demodulator.
- Keep Switch **SW1** in PWM position.
- Connect the output of PWM/PPM Demodulator post **DEMOD OUT** to post **IN** of Filter 1.
- Observe PWM signal at PWM OUT Post. Variation in width of square wave is high because the frequency is high. Due to persistence of vision, only blurt band in the waveform will be observed. If the modulating frequency is kept low around 1-10Hz by keeping FG-02 jumper **JP4** in **1-10Hz** range, we can observe the variation in the width of square wave. If the signal generator is OFF, only square wave of fundamental frequency and fixed ON time will be observed and no width variations are present.
- Observe the received signal at Filter O/P post **OUT** Demodulated signal represents original signal.

SWITCH FAULTS

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1 (SF1)** in Switch Fault section to ON position. This will open resistor R24 from the feedback loop of PWM modulator driver stage. U20 will operate in open loop mode & o/p gets distorted.
- Put switch **4 (SF1)** in Switch Fault section to ON position. This will open capacitor C223 & C218 from first stage of Filter 2. Filter o/p gets distorted.
- Put switch **5 (SF2)** in Switch Fault section to ON position. This will short Capacitor C258 & C242 at PPM demodulator first stage & duty cycle of waveform will change.
- Put switch **6 (SF2)** in Switch Fault section to ON position. This will short Capacitor C85 & C244 at PPM demodulator final stage. This will cause distortion in integrator response.
- Put switch **7 (SF2)** in Switch Fault section to ON position. This will short Capacitor C84 & C241 at PPM modulator stage. This will cause change in width of the pulse.

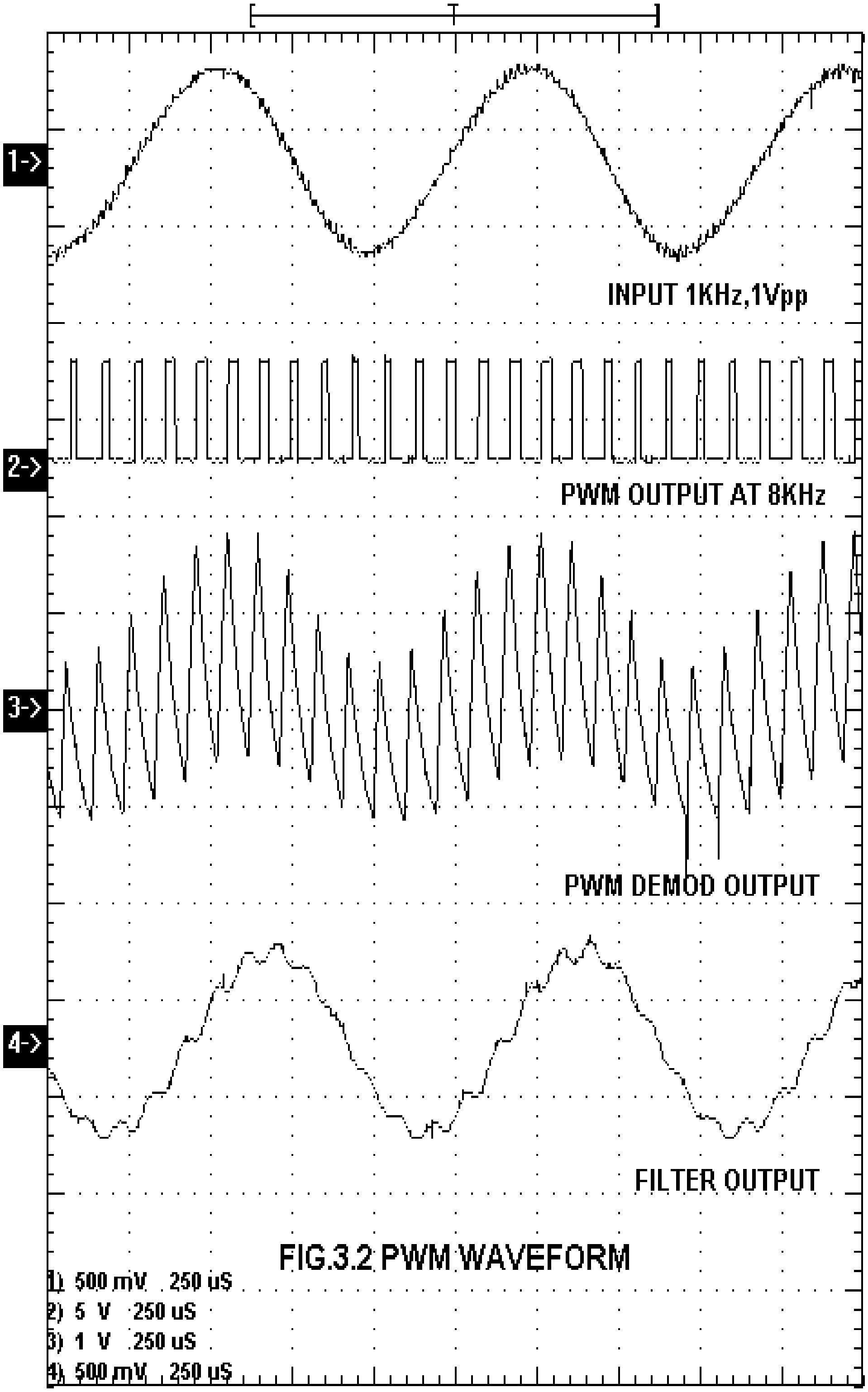


FIG.3.2 PWM WAVEFORM

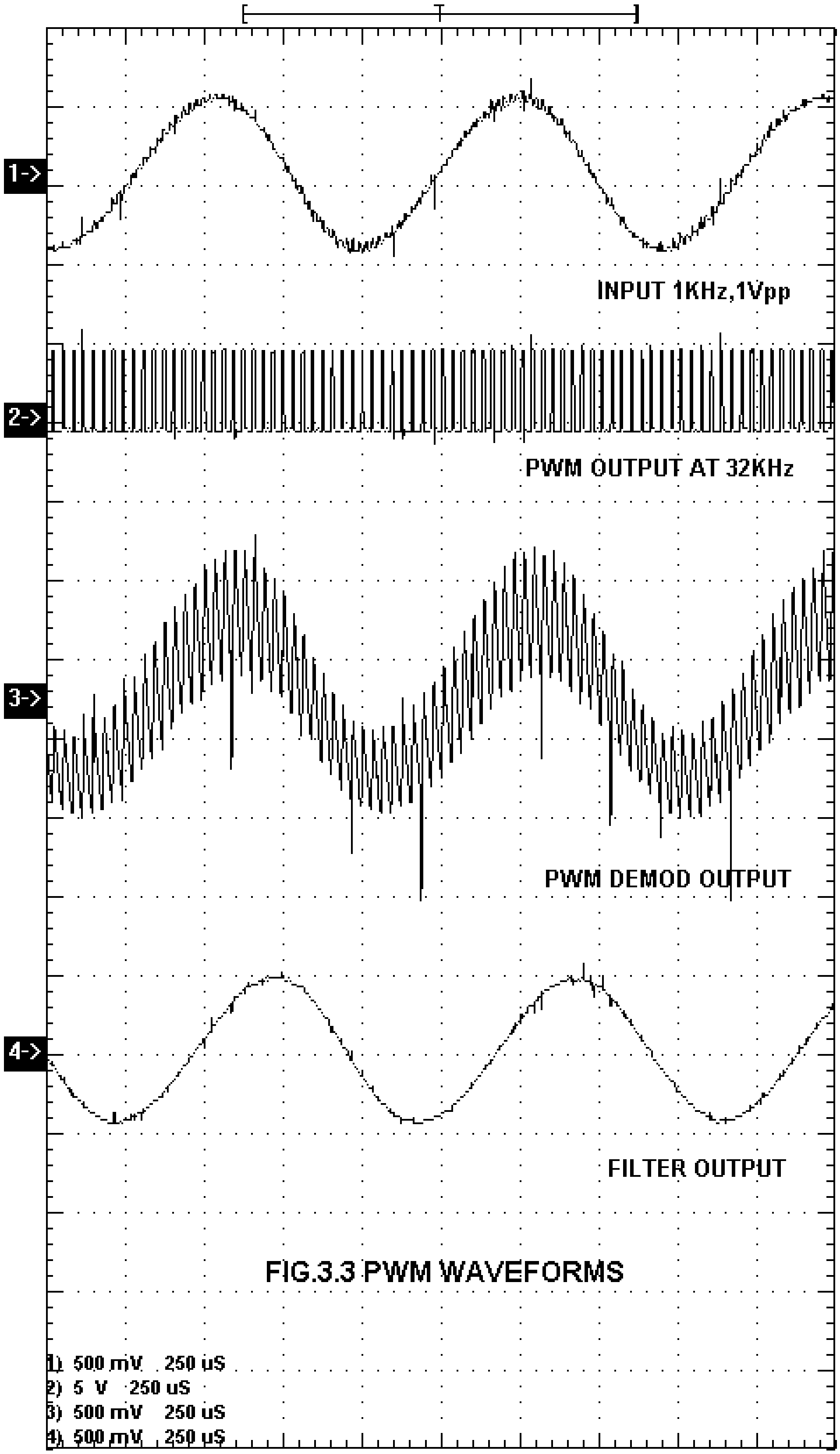
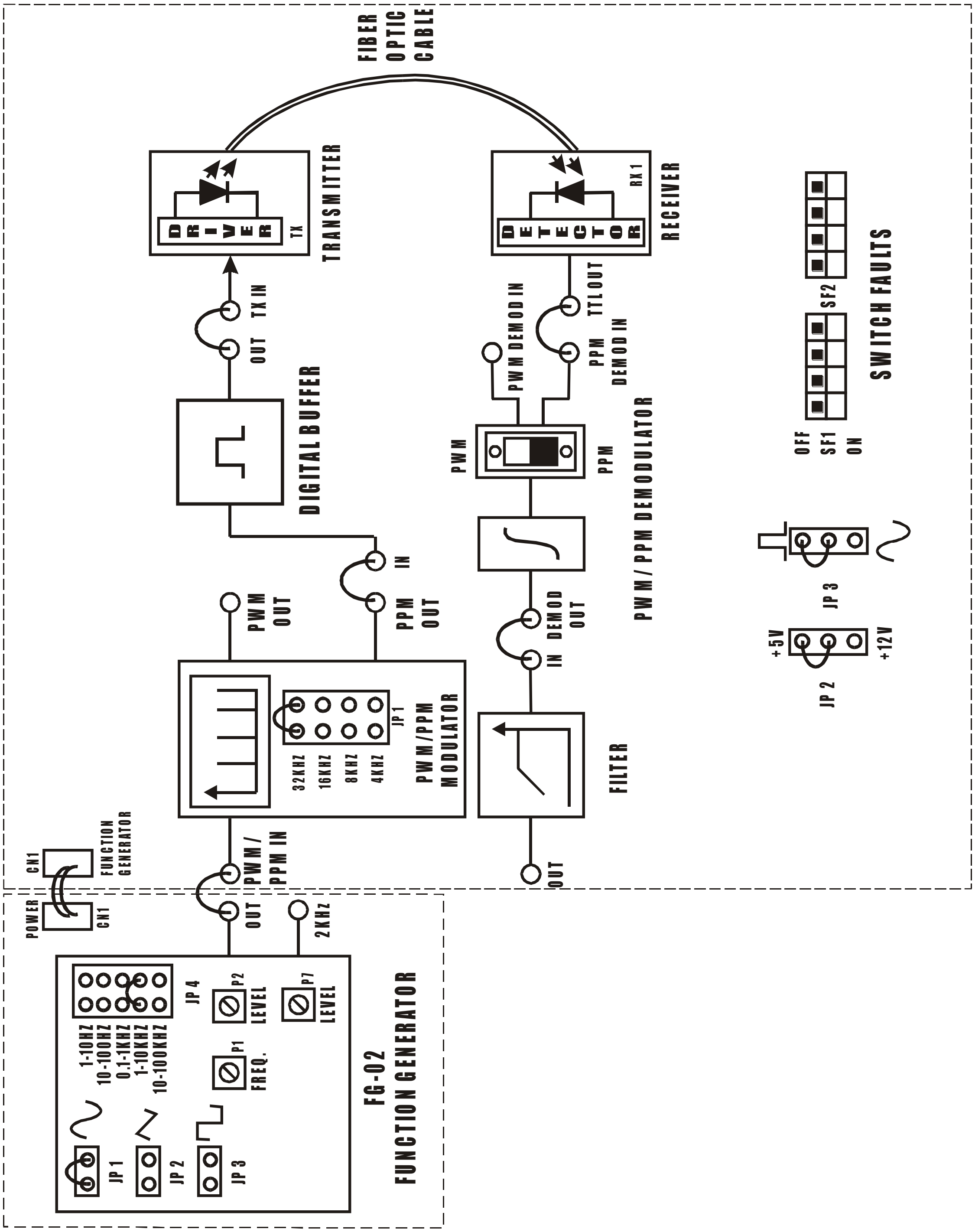


FIG.3.3 PWM WAVEFORMS

EXPERIMENT NO.4



FCL-03

FIG.4.1 BLOCK DIAGRAM FOR PULSE POSITION MODULATION AND DEMODULATION

EXPERIMENT NO.4

NAME

Study of Pulse Position Modulation and Demodulation

OBJECTIVE

The objective of this experiment is to study the circuit action of Pulse Position Modulation and Demodulation over Fiber Optic Digital Link.

THEORY

The position of the TTL pulse is changed on time scale according to the variation of input modulating signal amplitude. Pulse width modulated signal is fed as input to this circuit. Please note that input modulating signal must be converted into pulse width modulated form before applying to pulse position modulator. As the signal is PWM, naturally according to the input signal, the pulse duration is changing and this change in pulse duration causes for the delay in triggering. The input is given to trailing edge trigger input of monoshot. So finally we get the pulses at the output, which are shifted on the time slot. This is nothing but pulse position modulation.
Thus Pulse Positions are directly proportional to the instantaneous values of modulating signal. This PPM signal is then transmitted through transmitter (SFH756v) and received at detector output (SFH551v).

EQUIPMENTS

- FCL-03
- FG-02 with power cable
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position

PROCEDURE

- Make connections as shown in fig.4.1. Connect the power supply cables with proper polarity to FCL-03 Kit. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-02 to FCL-03 using power cable.
- Switch on the power supply.
- Keep the jumpers **JP1**, **JP2** & **JP3** on FCL-03 as shown in fig.4.1.
- Keep the function generator **JP1** shorted & **JP2**, **JP3** open.
- Keep the function generator **JP4** in **1-10KHz** position.
- Connect the **OUT** Signal from FG-02 to the **PWM/PPM IN** post of PWM/PPM Modulator on FCL-03 and keep the signal frequency at **1KHz** & Amplitude at **1Vpp**.
- Connect the **PPM OUT** Signal from PWM/PPM Modulator to the **IN** post of Digital Buffer on FCL-03.
- Connect the output of Digital Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of **RX1** Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect the output of detector post **TTL OUT** to post **PPM DEMOD IN** of PWM/PPM Demodulator.
- Keep Switch **SW1** in PPM position.
- Connect the output of PWM/PPM Demodulator post **DEMOD OUT** to post **IN** of Filter 1.
- Observe PPM signal at PPM OUT Post. Variation in position of square wave is high because the frequency is high. Due to persistence of vision, only blurt band in the waveform will be observed. If the modulating frequency is kept low around 1-10Hz by keeping FG-02 jumper **JP4** in **1-10Hz** range, we can observe the variation in the position of square wave. If the signal generator is OFF, only square wave of fundamental frequency and fixed position will be observed and no position variations are present.
- Observe the received signal at Filter O/P post **OUT** Demodulated signal represents original signal.

SWITCH FAULTS

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1 (SF1)** in Switch Fault section to ON position. This will open Resistor R24 from the feedback loop of PWM modulator driver stage. U20 will operate in open loop mode & o/p gets distorted.
- Put switch **4 (SF1)** in Switch Fault section to ON position. This will open capacitorC223 & C218 from first stage of Filter 2. Filter o/p gets distorted.
- Put switch **6 (SF2)** in Switch Fault section to ON position. This will short Capacitor C85 & C244 at PPM demodulator final stage. This will cause distortion in integrator response.
- Put switch **7 (SF2)** in Switch Fault section to ON position. This will short Capacitor C84 & C241 at PPM modulator stage. This will cause change in width of the pulse.

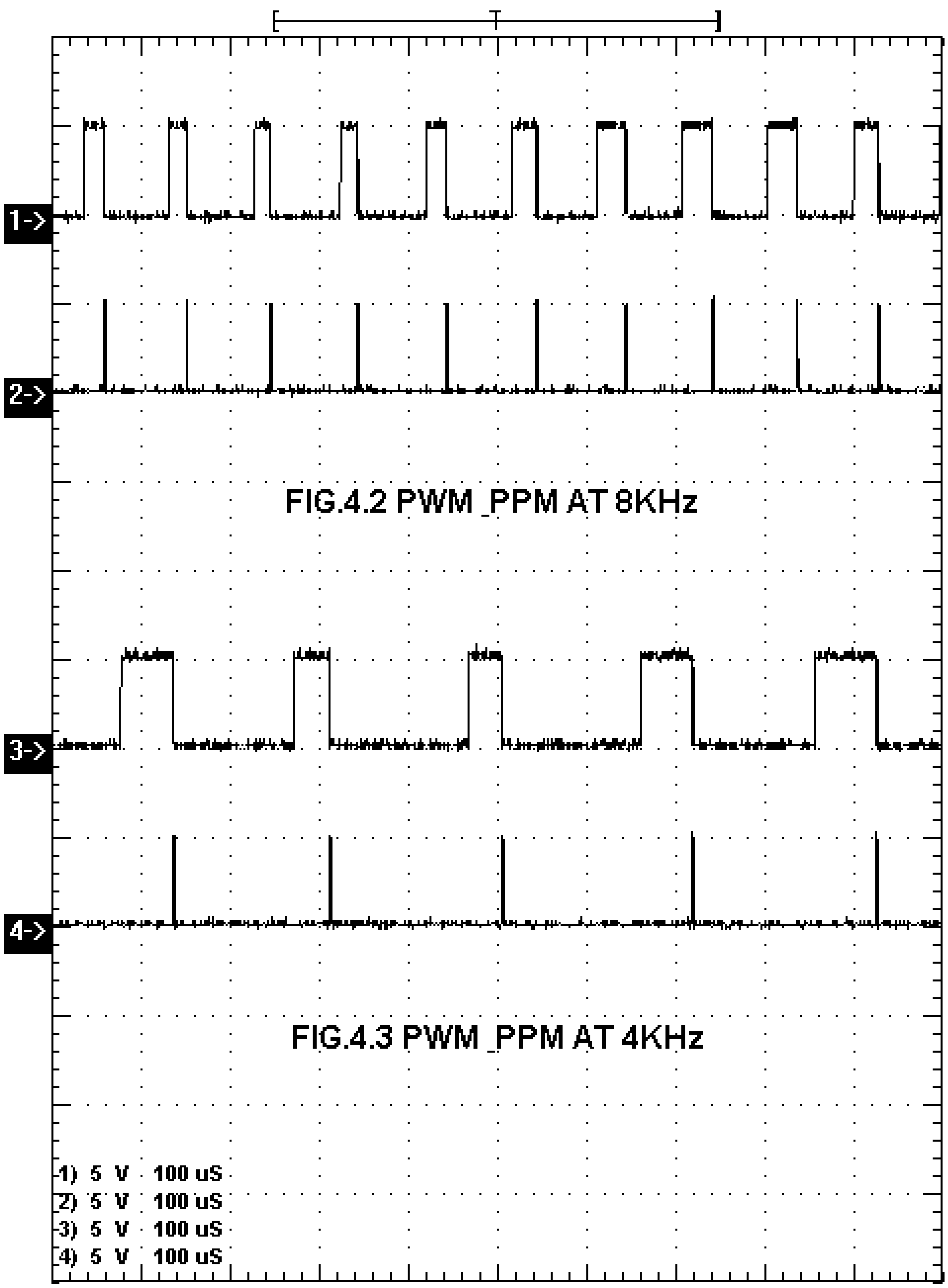


FIG.4.2 PWM _PPM AT 8KHz

FIG.4.3 PWM _PPM AT 4KHz

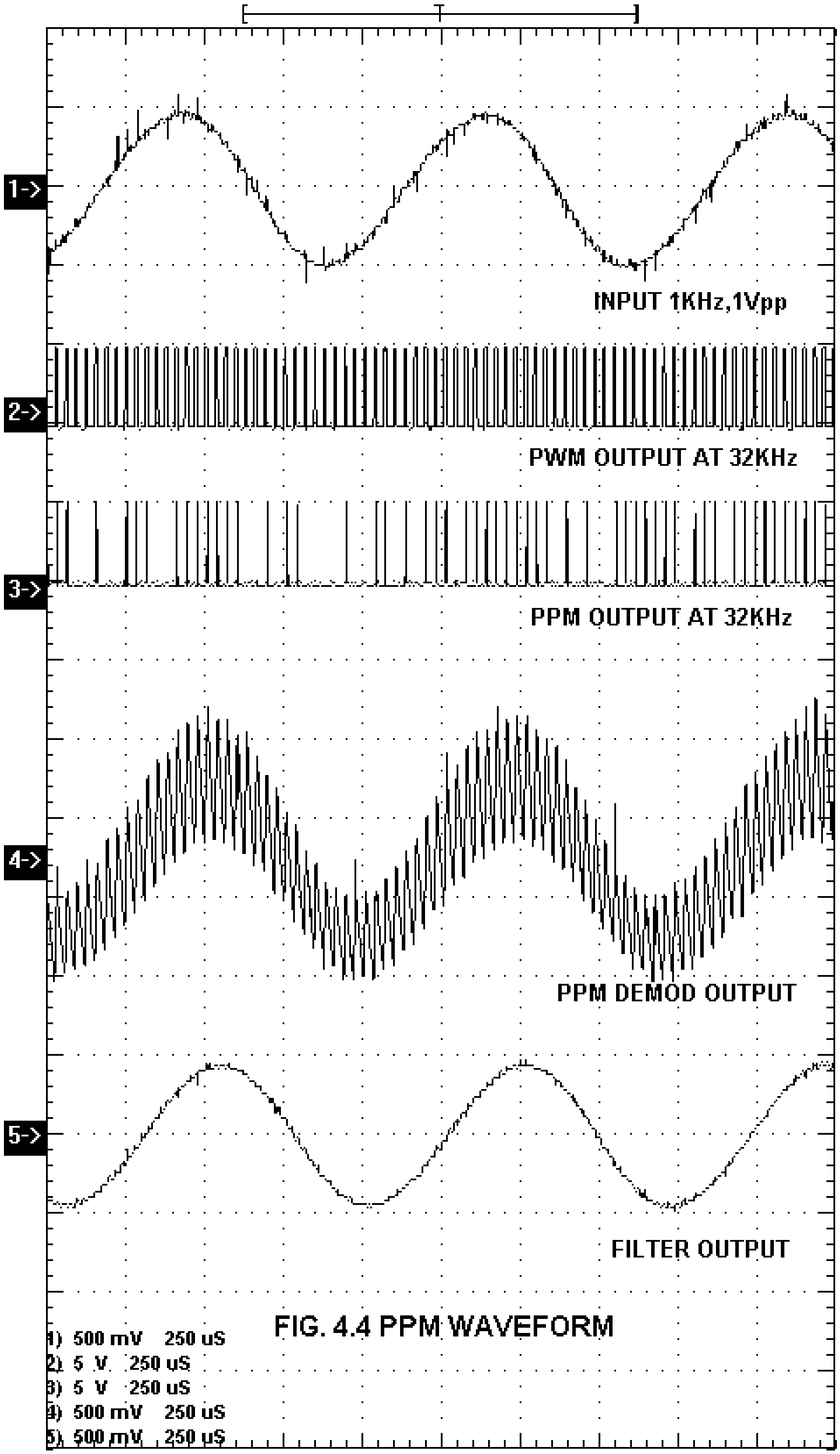


FIG. 4.4 PPM WAVEFORM

EXPERIMENT NO.5

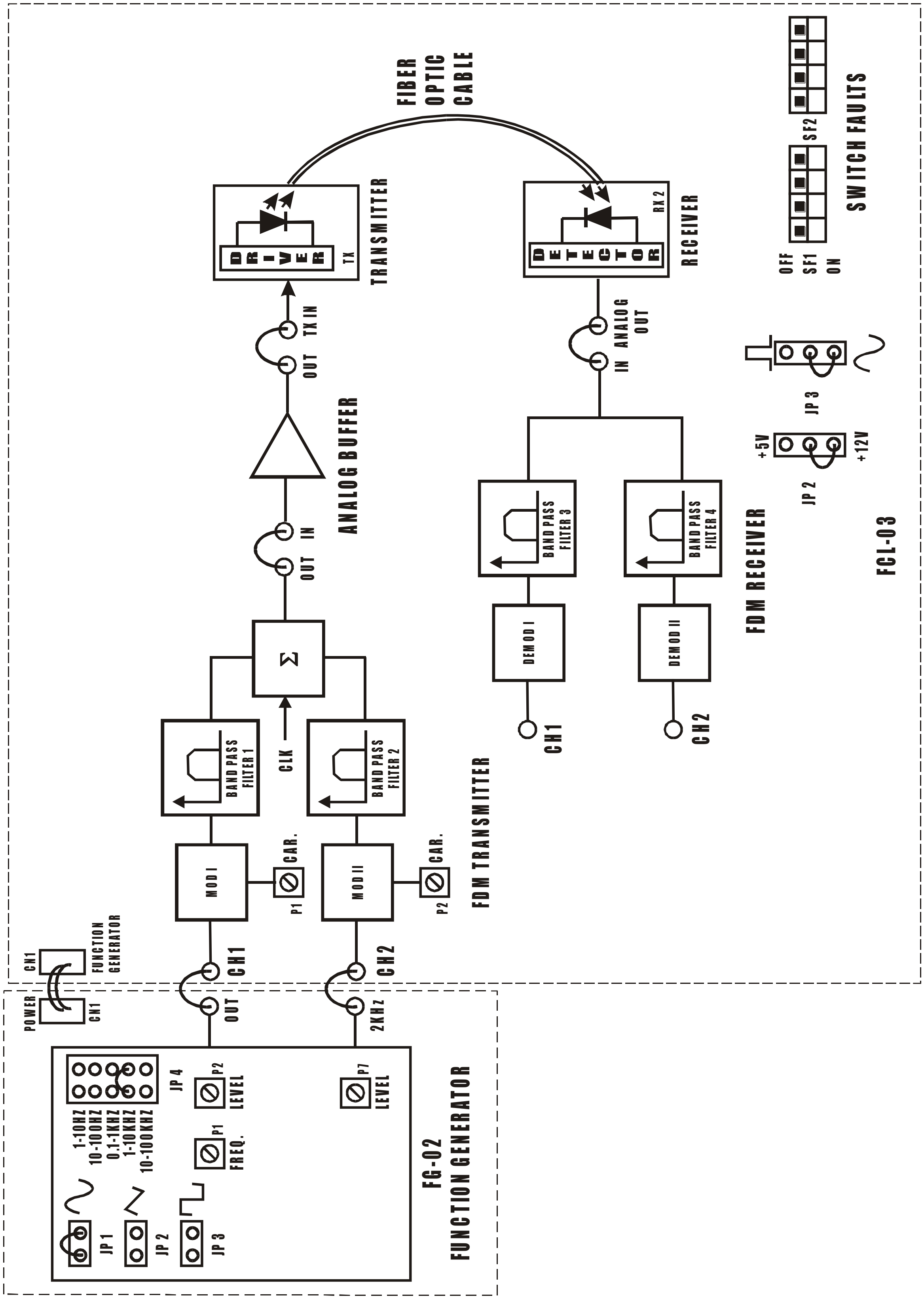


FIG. 5.1 BLOCK DIAGRAM FOR FREQUENCY DIVISION MULTIPLEXING AND DEMULTIPLEXING

EXPERIMENT NO.5

NAME

Study of Frequency Division Multiplexing And Demultiplexing

OBJECTIVE

The objective of this experiment is to study Frequency Division Multiplexing & Demultiplexing over Fiber Optic Analog Link.

THEORY

Multiplexing is a signal processing operation by which a number of independent messages can be combined into a composite signal suitable for transmission over common channel. The technique of separating these messages in frequency is referred to as frequency division multiplexing (FDM).

The incoming signal is first passed through a low pass filter to remove high frequency components that do not contribute significantly to signal representation but disturb other messages. The filtered signal is then applied to modulators that shift the frequency ranges of the signals so as to occupy mutually exclusive frequency intervals. The necessary carrier frequencies needed to perform these frequency translations are obtained from a carrier supply. Then input signal is modulated with respect to carrier signal. Modulation technique used here is frequency modulation technique. The band pass filters following the modulators are used to respect the band of each modulated wave to its prescribed range. The resulting band pass filter outputs are combined in parallel to form the input to the common channels.

Message recovery or demodulation of FDM is accomplished as follows:

The received signal is applied to bank of band pass filters with their input connected in parallel. This technique is used to separate message signals on frequency Occupancy basis. Finally the original message signals are recovered by individual demodulators.

Note that FDM operates in only one direction. To achieve two way transmission, as in telephony, for e.g. we have to completely duplicate the demultiplexing facilities with components connected in reverse order & with the signal waves preceding from right to left.

EQUIPMENTS

- FCL-03
- FG-02 with power cable
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.5.1. Connect the power supply cables with proper polarity to FCL-03 Kit. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-02 to FCL-03 using power cable.
- Keep the jumpers **JP2** & **JP3** on FCL-03 as shown in fig.5.1.
- Switch on the power supply.
- Keep the function generator **JP1** shorted & **JP2**, **JP3** open.
- Keep the function generator **JP4** in **1-10KHz** position.
- Connect the **OUT** Signal from FG-02 to the **CH1** post of FDM Transmitter on FCL-03 and keep the signal frequency at **1KHz** & Amplitude at **0.5Vpp**
- Connect the **2KHz, 0.5Vpp** Signal from FG-02 as a constant signal to the **CH2** post of FDM Transmitter on FCL-03.
- Connect the **OUT** post of FDM Transmitter to the **IN** post of Analog Buffer on FCL-03.
- Then set carrier frequency of MOD I of about 10KHz with the help of **P1** and carrier freq. Of MOD II of about 20KHz with the help of pot **P2**.
- Then observe the signal at Band Pass Filter-1 **OUT** it should be maximum signal transmission at center freq. fc of about 10KHz.
- Then observe the signal at Band Pass Filter-2 **OUT** it should be maximum signal transmission at center freq. fc of about 20KHz.
- Then observe signal at out post of summation block as shown in waveform Fig. 5.2.
- Connect the output of Analog Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Now rotate the Optical Power Control pot **P3** in FCL-03 in anticlockwise direction. This ensures minimum current flow through LED.

- Slightly unscrew the cap of **RX2** Photo Diode SFH250V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Observe the output signal from the detector at **ANALOG OUT** post on Oscilloscope by adjusting Optical Power Control Pot **P3** and you should get the reproduction of the original transmitted signal.
- Connect the output of detector post **ANALOG OUT** to post **IN** of FDM Receiver.
- Observe the demultiplexed & demodulated signal at **CH1** & **CH2** output posts of FDM Receiver, fine tune carrier frequency of modulator so as it will give clear reproduction of input signal as shown in Fig.5.3.
- Connect CH1 & CH2 post to IN post of filter 1 & filter 2 respectively.
- Observe the signal at OUT post of filter 1 & filter 2 as a reproduction of original signal.

SWITCH FAULTS

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **2 (SF1)** in Switch Fault section to ON position. This will open Capacitor C51 from second stage of Band Pass Filter 3 & center freq. Of BPF no.3 gets distorted.
- Put switch **3 (SF1)** in Switch Fault section to ON position. This will open Resistor R47 in series with Potentiometer P14 from the FM demodulator 2. This will result in change of free running freq.
- Put switch **4 (SF1)** in Switch Fault section to ON position. This will open Capacitor C223 & C218 from first stage of Filter 2. Filter o/p gets distorted.
- Put switch **8 (SF2)** in Switch Fault section to ON position. This will short Resistor R135 & R168 from the Band Pass Filter 1.This will result in distorted Output.

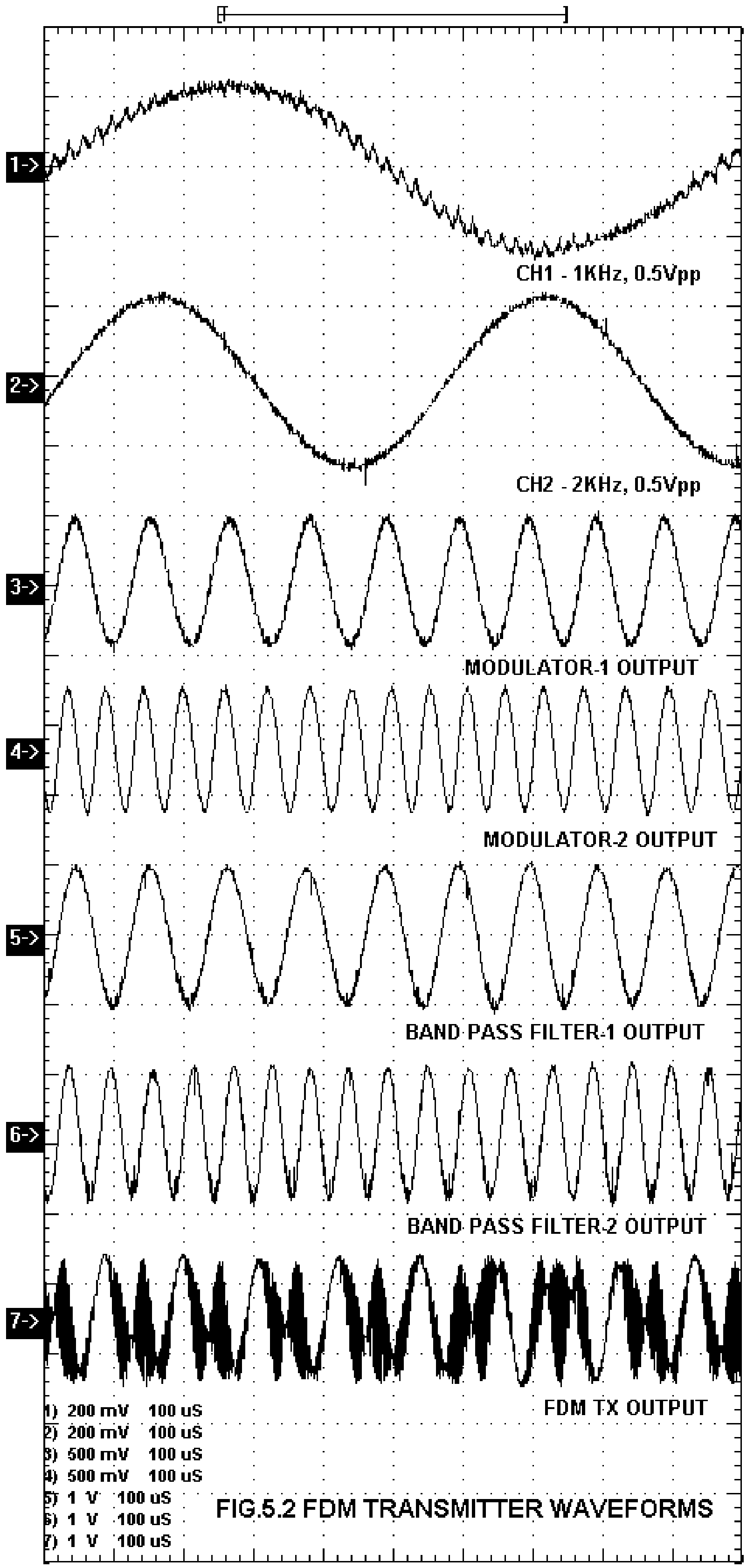


FIG.5.2 FDM TRANSMITTER WAVEFORMS

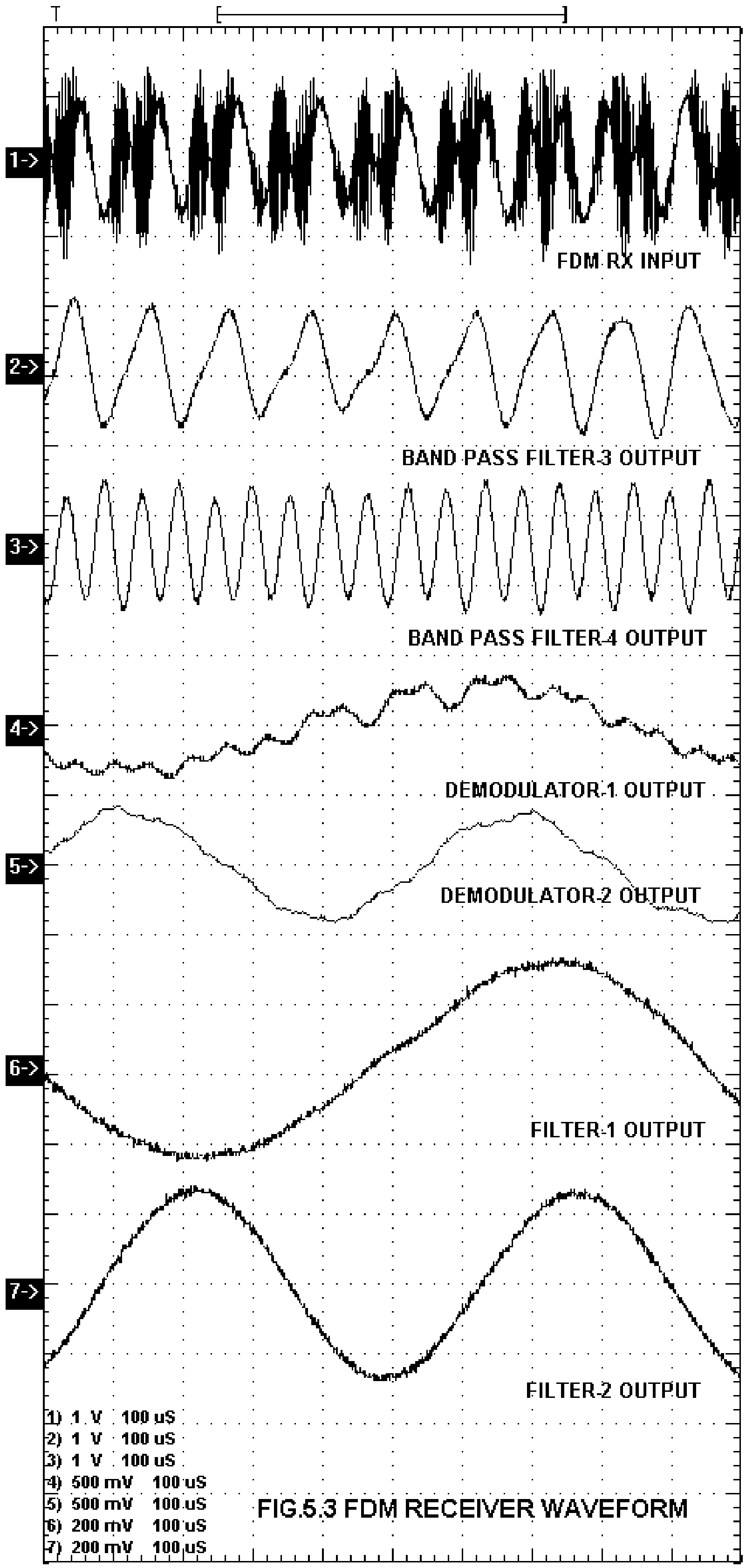


FIG.5.3 FDM RECEIVER WAVEFORM